

This lecture note was done by Choongrak Kim(Professor, Department of Statistics, Pusan National University), based on “Regression Analysis” (Authors : Choongrak Kim and Geunseok Kang ; Publisher : Gyowoo Sa, 2nd edition) which is written in Korean. In fact, this note is an English version of the text.

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January 1, 2019
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2019년 1월 1일
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APPENDIX. Matrix Theory

A.1 Basic Theories

A.1.1 Types of Matrix

(1) The $m \times n$ matrix A with the ij component a_{ij} is given by

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

and denoted by $A = (a_{ij})$

(2) The i th row vector of A : $\mathbf{a}_i = (a_{i1} \ a_{i2} \ \cdots \ a_{in})$

The j th column vector of A :

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

(3) square matrix : $m = n$ case

(4) diagonal matrix : a square matrix with $a_{ij} = 0, i \neq j$, and denoted by

$$A = \text{diag}(a_1, \cdots, a_n)$$

(5) unit matrix or identity matrix : a diagonal matrix with $a_{ii} = 1, i = 1, \cdots, n$,

and denoted by I or I_n

(6) The *transpose* of an $m \times n$ matrix $A = (a_{ij})$ is an $n \times m$ matrix $A' = (a_{ji})$, and also denoted by A^T or A^t . Further, it is clear that $(AB)' = B'A'$.

(7) A square matrix A is called *idempotent* if $A^2 = A$.

A.1.2 Trace

(1) The *trace* of a square matrix A is sum of diagonal elements of A , and denoted by $\text{tr}(A)$. Hence, $\text{tr}(A) = \sum a_{ii}$.

(2) properties of trace

$$\text{tr}(A \pm B) = \text{tr}(A) \pm \text{tr}(B)$$

$$\text{tr}(kA) = k \text{tr}(A), \quad k : \text{constant}$$

$$\text{tr}(AB) = \text{tr}(BA),$$

where AB and BA should be defined.

A.1.3 Determinant

(1) The *determinant* of an $n \times n$ square matrix A is denoted by $\det(A)$ or $|A|$, and defined as

$$\det(A) = \sum_{j=1}^n a_{ij}c_{ij}, \quad \forall i = 1, \dots, n,$$

where

$$c_{ij} = (-1)^{i+j}d_{ij}$$

is called the *cofactor* of a_{ij} and d_{ij} is the determinant of $(n-1) \times (n-1)$ matrix $A_{(i,j)}$ which is matrix A with the i th row and j th column deleted.

(2) Examples

$n = 1$ case : $|A| = a_{11}$, i.e, the value itself

$n = 2$ case :

$$\begin{aligned} |A| &= \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}, \quad (i = 1 \text{ case}) \\ &= -a_{21}a_{12} + a_{22}a_{11}, \quad (i = 2 \text{ case}) \end{aligned}$$

$n = 3$ case :

$$\begin{aligned} |A| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\ &= a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} \\ &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}a_{22}a_{33} - a_{11}a_{32}a_{23} - a_{12}a_{21}a_{33} + a_{12}a_{31}a_{23} \\ &\quad + a_{13}a_{21}a_{32} - a_{13}a_{31}a_{22} \end{aligned}$$

(2) properties of determinant

$$|AB| = |BA| = |A||B|, \quad \text{if } A \text{ and } B \text{ are } n \times n \text{ square matrix}$$

$$|A| = |A'|$$

$$|kA| = k^n |A|, \quad k \text{ is constant}$$

(3) A square matrix A is called *non-singular* if $|A| \neq 0$, and called *singular* if $|A| = 0$.

A.2 Inverse Matrix

A.2.1 Linearly Independence and Dependence

(1) Def : The *linear combination* of n -dimensional vector $v_1, \dots, v_k$ and constants $c_1, \dots, c_k$ is $\mathbf{0}$, i.e.,

$$c_1 v_1 + \dots + c_k v_k = \mathbf{0}$$

has at least one c_i which is nonzero, then $v_1, \dots, v_k$ are called *linearly dependent*.

Otherwise, if all the c_i 's are zero, then $v_1, \dots, v_k$ are called *linearly independent*.

(2) Example : Two vectors $(1, 1)$ and $(-3, 2)$ are linearly independent because

$$c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

gives $c_1 - 3c_2 = 0$, $c_1 + 2c_2 = 0$, and we must have $c_1 = 0$, $c_2 = 0$.

(3) Remark : If $v_1, \dots, v_k$ are linearly dependent, $c_1 v_1 + \dots + c_k v_k = \mathbf{0}$ has nonzero c_i . Assume that if $c_j \neq 0$, then

$$v_j = -\frac{1}{c_j}(c_1 v_1 + \dots + c_{j-1} v_{j-1} + c_{j+1} v_{j+1} + \dots + c_k v_k)$$

i.e., if $v_1, \dots, v_k$ are linearly dependent, then one vector can be expressed as a linear combination of other vectors.

A.2.2 Rank of Matrix

(1) Def : Let A be an $m \times n$ matrix with m row vectors and n column vectors. Let m^* be the maximum number of linearly independent vectors among m row vectors, and n^* be the maximum number of linearly independent vectors among n column vectors. Then, we must have $m^* = n^*$ which is called the *rank* of A , and denoted by $r(A)$. Hence, $r(A) \leq \min(m, n)$.

(1) Example : Compute the rank of 5×4 matrix A .

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 1 & -1 & 3 & 2 \\ 3 & 0 & 6 & 5 \\ 2 & 1 & 3 & 3 \\ 6 & 0 & 12 & 10 \end{bmatrix}$$

First, we must have $r(A) \leq 4$. Note that the 1st and 2nd row vectors are linearly independent, and the 3rd, 4th, and 5th row vectors can be expressed as a linear combination of the 1st and 2nd row vectors. For, example, the 3rd row vector can be expressed as $(1, 2, 0, 1) + 2(1, -1, 3, 2)$. Also, can show that the 1st and 2nd column vectors are linear independent. Therefore, $r(A) = 2$.

(2) properties of rank

$$(I) \ r(\mathbf{AB}) \leq \min(r(\mathbf{A}), r(\mathbf{B}))$$

(II) If $n \times n$ matrix $\mathbf{A}$ is non-singular, then $r(\mathbf{A}) = n$, and if it is singular, then $r(\mathbf{A}) < n$.

(III) The rank of $\mathbf{A}$ does not change if it is multiplied by a non-singular matrix.

(IV) If $\mathbf{AGA} = \mathbf{A}$, then $r(\mathbf{A}) = r(\mathbf{GA})$.

(V) $r(\mathbf{A:B}) \leq r(\mathbf{A}) + r(\mathbf{B})$, where $\mathbf{A}$ is $m \times n_1$, $\mathbf{B}$ is $m \times n_2$, and $\mathbf{A:B}$ is $m \times (n_1 + n_2)$ matrix, called *augmented matrix*. For example, when

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 7 & 9 \\ 8 & 10 \end{pmatrix}$$

$$\mathbf{A:B} = \begin{pmatrix} 1 & 3 & 5 & 7 & 9 \\ 2 & 4 & 6 & 8 & 10 \end{pmatrix}$$

$$(VI) \ r(\mathbf{A} + \mathbf{B}) \leq r(\mathbf{A:B}) \leq r(\mathbf{A}) + r(\mathbf{B})$$

(VII) If $\mathbf{A}$ is $n \times n$ matrix, then $r(\mathbf{AB}) \geq r(\mathbf{A}) + r(\mathbf{B}) - n$.

(VIII) If $\mathbf{A}$ is $n \times n$ idempotent matrix, then $r(\mathbf{I} - \mathbf{A}) = n - r(\mathbf{A})$.

$$(IX) \ r(\mathbf{A}) = r(\mathbf{A}') = r(\mathbf{A}'\mathbf{A}) = r(\mathbf{AA}')$$

A.2.3 Inverse Matrix

(1) motivation : To compute a solution x of the equation $ax = b$, we multiply the inverse of a , i.e., $1/a$ on both sides, i.e.,

$$ax = b \Rightarrow \frac{1}{a} ax = \frac{1}{a} b \Rightarrow x = \frac{b}{a}$$

Here, we must have $a \neq 0$ since the existence of the inverse of a is possible only when $a \neq 0$. Now, consider n equations with n unknowns.

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Now, let $A = (a_{ij})$, $i = 1, \dots, n$; $j = 1, \dots, n$ be $n \times n$ square matrix, $\mathbf{x} = (x_1 \ x_2 \ \cdots \ x_n)'$, $\mathbf{b} = (b_1 \ b_2 \ \cdots \ b_n)'$, then the above equations can be expressed as

$$A\mathbf{x} = \mathbf{b}$$

If A^{-1} satisfies

$$AA^{-1} = A^{-1}A = I$$

then it is called the *inverse* of A . For a scalar a to be invertible, we must have $a \neq 0$. Likewise, for a square matrix A to be invertible, we must have

$|A| \neq 0$, i.e., non-singular. Hence, if A is non-singular, then $A^{-1}Ax = A^{-1}b \Rightarrow x = A^{-1}b$. In fact, A^{-1} is given by

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} c_{11} & c_{21} & \cdots & c_{n1} \\ \vdots & \vdots & & \vdots \\ c_{1n} & c_{2n} & \cdots & c_{nn} \end{bmatrix}$$

where c_{ij} is the cofactor (see A.1.3) of a_{ij} .

(2) properties

(I) A^{-1} is unique.

(II) A^{-1} is non-singular.

(III) $(A')^{-1} = (A^{-1})'$

(IV) $(AB)^{-1} = B^{-1}A^{-1}$

(3) orthogonal matrix

If a square matrix A satisfies $A'A = AA' = I$, then it is called *orthogonal* matrix, and it has the following properties.

(I) $\text{tr}(A'BA) = \text{tr}(B)$

(II) $|A'BA| = |B|$

(III) $|A| = \pm 1$

A.2.4 Inverse of Special Matrices

(I) Let J be an $n \times n$ square matrix with all the components are 1, then

if $a \neq 0, a + nb \neq 0$, we have

$$(aI + bJ)^{-1} = \frac{1}{a} \left(I - \frac{b}{a + nb} J \right)$$

(II) $\{\text{diag}(a_1, \dots, a_n)\}^{-1} = \text{diag}(1/a_1, \dots, 1/a_n), a_i \neq 0, i = 1, \dots, n$

(III) $I + A + A^2 + \dots + A^{n-1} = (A^n - I)(A - I)^{-1}$

(IV) $(I + A^{-1})^{-1} = A(A + I)^{-1}$

(V) $(I + AB)^{-1} = I - A(I + BA)^{-1}B$

(VI) $(A + UBV)^{-1} = A^{-1} - A^{-1}UBV(I + A^{-1}UBV)^{-1}A^{-1}$

A.2.5 Generalized Inverse

Here we define the inverse of A when A is non-singular, and we also define the inverse of A when A is not a square matrix.

(1) Moore-Penrose inverse

Let A be $p \times q$ matrix, then the *Moore – Penrose* inverse of A is a $q \times p$ matrix M satisfying the following 3 conditions;

(I) $AMA = A$

(II) $MAM = M$

(III) AM and MA are symmetric

Note that the Moore-Penrose inverse M is unique. For example, the Moore-Penrose inverse of for 3×4 matrix

$$A = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 2 & 2 & 2 \\ -1 & 4 & 5 & 3 \end{bmatrix}$$

is

$$M = \frac{1}{18} \begin{bmatrix} 5 & 2 & -1 \\ 1 & 1 & 1 \\ -4 & -1 & 2 \\ 6 & 3 & 0 \end{bmatrix}$$

Also, note that if A is a square matrix, then M is also a square matrix and $M = A^{-1}$.

(2) generalized inverse

Among the above 3 conditions, if G satisfies the 1st condition only, i.e., $AGA = A$, then G is called the *generalized inverse* of A , and denoted by A^- . In fact, G is not unique. Here, we introduce one popular method of computing G . First, find a square and non-singular submatrix A_{11} of $A_{p \times q}$, and let all other components be $\mathbf{0}$. Finally, find A_{11}^{-1} , i.e.,

$$A_{p \times q} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \Rightarrow G_{q \times p} = \begin{bmatrix} A_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$$

For example, consider

$$A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 3 & -1 & 2 & -2 \\ 5 & -4 & 0 & -7 \end{bmatrix}$$

and let

$$A_{11} = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}, A_{12} = \begin{bmatrix} 4 & 3 \\ 2 & -2 \end{bmatrix}, A_{21} = \begin{bmatrix} 5 & -4 \end{bmatrix}, A_{22} = \begin{bmatrix} 0 & -7 \end{bmatrix}$$

then

$$G = \frac{1}{7} \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

As an another example, consider a oneway classification

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, 2, \quad j = 1, 2, 3$$

i.e.,

$$\begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{13} \\ Y_{21} \\ Y_{22} \\ Y_{23} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \alpha_1 \\ \alpha_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{23} \end{bmatrix}$$

In matrix notation,

$$y = X\beta + \varepsilon$$

then the normal equation for the least squares method becomes

$$(X'X)\beta = X'y,$$

however,

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 6 & 3 & 3 \\ 3 & 3 & 0 \\ 3 & 0 & 3 \end{bmatrix}$$

is singular, so that $(\mathbf{X}'\mathbf{X})^{-1}$ does not exist. Hence, as an estimator of β , we use the generalized inverse $(\mathbf{X}'\mathbf{X})^{-}$, and the estimator becomes $(\mathbf{X}'\mathbf{X})^{-}\mathbf{X}'\mathbf{y}$.

One possible method is given by

$$(\mathbf{X}'\mathbf{X})^{-} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}$$

(3) computational issues in inverse matrix

Note that

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$$

is singular since $|\mathbf{A}| = 0$, and therefore the inverse does not exist. Now,

$$\mathbf{B} = \begin{bmatrix} 1.9998 & 0.9999 \\ 5.9994 & 3.0009 \end{bmatrix}$$

is non-singular since its determinant is 0.0024. Note that even though $\mathbf{B}$ is mathematically non-singular, it is almost singular since its determinant is close to 0. Now, consider

$$\begin{bmatrix} 2.0 & 2.5 \\ 2.5 & 3.0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.45 \\ 0.55 \end{bmatrix}$$

then the solution is $(0.1, 0.1)'$, however, the solution of

$$\begin{bmatrix} 2.04 & 2.49 \\ 2.49 & 3.04 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.45 \\ 0.55 \end{bmatrix}$$

is $(-1, 1)'$, and the determinant is very close to 0 (in fact, 0.0015). In this case, a small change in components may result in a big change in solution.

A matrix with a small value of determinant is called *ill – conditioned* matrix, and it can be measured by the *condition number* which is defined as a ratio of the largest singular value (see A.6.2 for definition) to the smallest singular value of the $n \times p$ matrix A . If a condition number is large, then the matrix can be ill-conditioned.

A.3 Partitioned Matrix

Consider partitioning the matrix P .

$$P = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Then, by the following identity,

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & 0 \\ C & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix}$$

we have

$$\begin{aligned} \left| \begin{bmatrix} A & B \\ C & D \end{bmatrix} \right| &= |A||D - CA^{-1}B|, \text{ if } A^{-1} \text{ exists} \\ &= |D||A - BD^{-1}C|, \text{ if } D^{-1} \text{ exists} \end{aligned}$$

Also, the inverse of P is

$$\begin{aligned}
& \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} \\
&= \begin{bmatrix} I & -A^{-1}B \\ 0 & I \end{bmatrix} \begin{bmatrix} A^{-1} & 0 \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{bmatrix}, \text{ if } A^{-1} \text{ exists} \\
&= \begin{bmatrix} 0 & 0 \\ 0 & D^{-1} \end{bmatrix} + \begin{bmatrix} I & \\ -D^{-1}C & \end{bmatrix} (A - BD^{-1}C)^{-1} [I - BD^{-1}], \text{ if } D^{-1} \text{ exists}
\end{aligned}$$

For example, if

$$P = \begin{bmatrix} 2 & 0 & 0 & 1 & -1 \\ 0 & 2 & 0 & 1 & -1 \\ 0 & 0 & 2 & 1 & -1 \\ 2 & 2 & 2 & 1 & 0 \\ -1 & -1 & -1 & 0 & 1 \end{bmatrix}$$

then let

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & -1 \end{bmatrix},$$

$$C = \begin{bmatrix} 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and we have

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned}
D - CA^{-1}B &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix} \cdot \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} -2 & 3 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}
\end{aligned}$$

so that

$$|P| = 8 \left(1 - \frac{9}{2}\right) = -28$$

$$P^{-1} = \frac{1}{14} \begin{bmatrix} 4 & -3 & -3 & 2 & -2 \\ -3 & 4 & -3 & 2 & -2 \\ -3 & -3 & 4 & 2 & -2 \\ 4 & 4 & 4 & 2 & 12 \\ -2 & -2 & -2 & 6 & 8 \end{bmatrix}$$

A.4 Eigenvalues and Eigenvectors

A.4.1 Eigenvalues and Eigenvectors

For a square matrix A , a vector $\mathbf{u}$, and a constant λ , consider

$$A\mathbf{u} = \lambda\mathbf{u}$$

which is equivalent to

$$(A - \lambda I)\mathbf{u} = \mathbf{0}.$$

If $A - \lambda I$ is non-singular, then the solution for $\mathbf{u}$ is $\mathbf{u} = \mathbf{0}$. But, if $A - \lambda I$ is singular, then there exist nonzero solution for $\mathbf{u}$ and a constant λ , i.e.,

$$|A - \lambda I| = 0$$

gives nonzero solution for $\mathbf{u}$ and a constant λ . The above equation is called the *characteristic equation* for A , and it is the n th degree polynomial in λ

since A is an $n \times n$. Let the n solutions be $\lambda_1, \dots, \lambda_n$, then these are called the *eigenvalue*, *characteristic roots*, or *latent roots*. Further, for each λ_i , a vector u_i satisfying

$$Au_i = \lambda_i u_i, \quad i = 1, \dots, n$$

is called the *eigenvector* corresponding to λ_i .

For example, if

$$A = \begin{bmatrix} 1 & 4 \\ 9 & 1 \end{bmatrix}$$

then the characteristic equation is

$$\begin{vmatrix} 1 - \lambda & 4 \\ 9 & 1 - \lambda \end{vmatrix} = 0$$

and eigenvalues are $\lambda = -5$ or $\lambda = 7$. Also, we have

$$A \begin{bmatrix} 2 \\ -3 \end{bmatrix} = -5 \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$A \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 7 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

and hence, the corresponding eigenvectors for $\lambda = -5$ and $\lambda = 7$ are

$$\begin{bmatrix} 2 \\ -3 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 3 \end{bmatrix},$$

respectively.

A.4.2 Properties of Eigenvalues

(I) If λ is an eigenvalue of A , then

- (a) Eigenvalue of A^k is λ^k .
- (b) Eigenvalue of A^{-1} is $1/\lambda$.
- (c) Eigenvalue of cA is $c\lambda$.
- (d) Eigenvalue of $A + cI$ is $\lambda + c$.
- (e) Eigenvalue of $(A + cI)^{-1}$ is $1/(\lambda + c)$.

(II) $\text{tr}(A) = \sum \lambda_i, |A| = \prod \lambda_i$

(III) If A is symmetric and its components are real values, then

- (a) All the eigenvalues are real.
- (b) Eigenvectors are orthogonal.
- (c) The rank of A is the number of nonzero eigenvalues.

(IV) Eigenvalues of idempotent matrix is either 0 or 1, but the converse is not true.

A.5 Quadratic Forms and Positive Definite Matrix

A.5.1 Quadratic Forms

For vector $\mathbf{x}$ and matrix A , $\mathbf{x}'A\mathbf{x}$ is called the *quadratic form* in $\mathbf{x}$. For example, if $\mathbf{x} = (x_1 \ x_2 \ x_3)'$ and

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 7 & 6 \\ 2 & -2 & 5 \end{bmatrix}$$

then

$$\mathbf{x}'A\mathbf{x} = x_1^2 + 7x_2^2 + 5x_3^2 + (4 + 2)x_1x_2 + (2 + 3)x_1x_3 + (-2 + 6)x_2x_3$$

which is a quadratic function in x_1, x_2, x_3 . In general, $\mathbf{x} = (x_1, \dots, x_n)'$

and $A_{n \times n} = (a_{ij})$, then

$$\begin{aligned} \mathbf{x}'A\mathbf{x} &= \sum a_{ii}x_i^2 + \sum_{i \neq j} \sum a_{ij}x_i x_j \\ &= \sum a_{ii}x_i^2 + \sum_{i < j} \sum (a_{ij} + a_{ji})x_i x_j \end{aligned}$$

Since the coefficient of $x_i x_j$ is the sum of a_{ij} and a_{ji} , the corresponding matrix A is not unique, i.e.,

$$\begin{aligned} &x_1^2 + 7x_2^2 + 5x_3^2 + (4 + 2)x_1x_2 + (2 + 3)x_1x_3 + (-2 + 6)x_2x_3 \\ &= x_1^2 + 7x_2^2 + 5x_3^2 + (3 + 3)x_1x_2 + (3 + 2)x_1x_3 + (0 + 4)x_2x_3 \end{aligned}$$

Hence, if we take

$$B = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 7 & 0 \\ 2 & 4 & 5 \end{bmatrix}$$

then $x'Ax = x'Bx$. Therefore, when we define a quadratic form, we often assume A is symmetric. Then, the matrix corresponding to a certain quadratic form becomes unique. A symmetric matrix corresponding to the above quadratic form is

$$A = \begin{bmatrix} 1 & 3 & 5/2 \\ 3 & 7 & 2 \\ 5/2 & 2 & 5 \end{bmatrix}$$

and it is unique.

A.5.2 Positive Definite Matrix

(1) definition

It is clear that a scalar is either positive or negative, however, it is not possible to define a matrix is positive or negative. Hence, the sign of a matrix is defined using the quadratic form. Consider a matrix

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 5 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

and the corresponding quadratic form is

$$\begin{aligned} x'Ax &= 2x_1^2 + 5x_2^2 + 2x_3^2 + 4x_1x_2 + 2x_1x_3 + 2x_2x_3 \\ &= (x_1 + 2x_2)^2 + (x_1 + x_3)^2 + (x_2 + x_3)^2 \end{aligned}$$

is always positive except that all the x_1, x_2, x_3 are zero. In this case, we call A is *positive definite*. In general,

A is called positive definite (p.d.) if $\mathbf{x}'A\mathbf{x} > 0, \forall \mathbf{x} \neq \mathbf{0}$.

On the other hand, consider

$$A = \begin{bmatrix} 37 & -2 & -24 \\ -2 & 13 & -3 \\ -24 & -3 & 17 \end{bmatrix}$$

and its quadratic form is

$$\begin{aligned} \mathbf{x}'A\mathbf{x} &= 37x_1^2 + 13x_2^2 + 17x_3^2 - 4x_1x_2 - 48x_1x_3 - 6x_2x_3 \\ &= (x_1 - 2x_2)^2 + (6x_1 - 4x_3)^2 + (3x_2 - x_3)^2 \end{aligned}$$

and which can be 0 even though $\mathbf{x} \neq \mathbf{0}$ because $\mathbf{x} = (2 \ 1 \ 3)'$ gives $\mathbf{x}'A\mathbf{x} = 0$.

This matrix is called *positive semidefinite* (p.s.d.) matrix. In general,

A is called positive semidefinite (p.s.d.) if $\mathbf{x}'A\mathbf{x} \geq 0, \forall \mathbf{x}$ and $\mathbf{x}'A\mathbf{x} =$

0, for some $\mathbf{x}$.

Positive definite and positive semidefinite matrix is often called *nonnegative definite*

(n.n.d.). In this sense, *negative definite* (n.d.), *negative semidefinite* (n.s.d.),

and *nonpositive definite* (n.p.d.) can be defined. Further, a matrix A is

called *indefinite* if A cannot be classified into any kind of definiteness.

(2) properties

(I) If $A_{n \times n} = (a_{ij})$ is p.d., then

$$(a) \ r(A) = n$$

(b) $a_{ii} > 0, i = 1, \dots, n$

(c) $P'AP$ is p.d. for any $n \times n$ square matrix P .

(II) If $A_{n \times n} = (a_{ij})$ is p.s.d., then

(a) $r(A) < n$

(b) $a_{ii} \geq 0, i = 1, \dots, n$

(c) $P'AP$ is n.n.d. for any $n \times n$ square matrix P .

(III) The necessary and sufficient condition for a symmetric matrix $A_{n \times n}$

to be p.d. is

(a) There exists a full rank matrix $B_{n \times n}$ s.t. $B'B = A$

(b) All the eigenvalues of A are positive

(c) $a_{11} > 0, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0, \dots, |A| > 0$

(IV) The necessary and sufficient condition for a symmetric matrix $A_{n \times n}$

to be p.s.d. is

(a) There exists a matrix $B_{n \times n}$ s.t. $B'B = A$ with $r(A) < n$

(b) All the eigenvalues of A are greater than 0 or equal to 0, and at least one eigenvalues should be 0

(V) If the rank of $A_{m \times n}$ is $m(m < n)$, then

(a) $A'A$ is p.s.d.

(b) AA' is n.n.d.

(VI) If the rank of $A_{m \times n}$ is $r(r < m, r < n)$, then

(a) $A'A$ is p.s.d.

(b) AA' is n.n.d.

A.6 Projection and Decomposition of Matrix

A.6.1 Projection

(I) projection on vectors

If we project y onto a vector x , then it becomes cx , where $c = \frac{x'y}{x'x}$.

(II) projection on column space

Let $X = (\mathbf{1}, x_1, \dots, x_{p-1})$ be $n \times p$ matrix, where $x_i, i = 1, \dots, p-1$

is n -vectors. Then, the *column space* of $\mathbf{X}$ is defined as follows;

$$\begin{aligned}\mathcal{C}_{\mathbf{X}} &\equiv \text{span}\{\mathbf{1}, \mathbf{x}_1, \dots, \mathbf{x}_{p-1}\} \\ &= \{ \beta_0 \mathbf{1} + \beta_1 \mathbf{x}_1 + \dots + \beta_{p-1} \mathbf{x}_{p-1} \mid \beta_0, \dots, \beta_{p-1} \in R \} \\ &= \{ \mathbf{X}\boldsymbol{\beta} \mid \boldsymbol{\beta} \in R^p \}\end{aligned}$$

If we project vector $\mathbf{y}$ onto the column space $\mathcal{C}_{\mathbf{X}}$, then it becomes $\mathbf{H}\mathbf{y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, and it is called a *projection* matrix.

(III) Gram-Schmidt orthogonalization

Let $\mathbf{X} = (\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{p-1})$ be an $n \times p$ matrix with $\mathbf{x}_0 = \mathbf{1}$. Also, let $\Pi(\mathbf{x}|\mathbf{z})$ be the projection of vector $\mathbf{x}$ onto a vector $\mathbf{z}$, i.e.,

$$\Pi(\mathbf{x}|\mathbf{z}) = (\mathbf{z}'\mathbf{x}/\mathbf{z}'\mathbf{z})\mathbf{z}$$

Transform $\mathbf{X} = (\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{p-1})$ to $\mathbf{Z} = (\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_{p-1})$ s.t.

$$\begin{aligned}\mathbf{z}_0 &= \mathbf{x}_0 \\ \mathbf{z}_1 &= \mathbf{x}_1 - \Pi(\mathbf{x}_1|\mathbf{z}_0) \\ \mathbf{z}_2 &= \mathbf{x}_2 - \Pi(\mathbf{x}_2|\mathbf{z}_0) - \Pi(\mathbf{x}_2|\mathbf{z}_1) \\ &\vdots \\ \mathbf{z}_{p-1} &= \mathbf{x}_{p-1} - \Pi(\mathbf{x}_{p-1}|\mathbf{z}_0) - \Pi(\mathbf{x}_{p-1}|\mathbf{z}_1) - \dots - \Pi(\mathbf{x}_{p-1}|\mathbf{z}_{p-2})\end{aligned}$$

Then, we have $\mathcal{C}_X = \mathcal{C}_Z$, and note that $z_0, z_1, \dots, z_{p-1}$ are orthogonal to each other, and this transformation is known as *Gram – Schmidt orthogonalization*.

A.6.2 Decomposition of Matrix

(I) QR Decomposition

Def. An $n \times p$ matrix X can be written as a form of $X = QR$, where Q is $n \times p$ orthogonal matrix and R is $p \times p$ upper triangular matrix. This is called QR decomposition of X .

There are 3 methods of computing QR decomposition; (i) Gram-Schmidt process, (ii) Householder transformation, (iii) Givens rotation. Here we introduce the Gram-Schmidt process. Assume that $X = (x_0, x_1, \dots, x_{p-1})$ is transformed to $Z = (z_0, z_1, \dots, z_{p-1})$ by the Gram-Schmidt orthogonalization, and define

$$\gamma_{ij} = \begin{cases} x'_j z_i / z'_i z_i & , i < j \\ 1 & , i = j \\ 0 & , i > j \end{cases}$$

Now, let $\Gamma = (\gamma_{ij})$, then it is $p \times p$ upper triangular matrix, and we have $X = Z\Gamma$. Therefore, $X = QR$, where $Q = Z$ and $R = \Gamma$. QR decomposition is very useful when we compute the inverse of $X'X$

because $X'X = R'Q'QR = R'R$ and it is very easy to compute the inverse of $R'R$.

(II) Cholesky Decomposition

For a symmetric and p.d. matrix A , there exists an upper triangular matrix R s.t. $A = R'R$, and this result is called the *Cholesky decomposition*.

The computation of R for a given A is based on the following algorithm.

STEP 1. $r_{11} = a_{11}^{1/2}$, $r_{ij} = a_{1j}/r_{11}$, $j = 2, \dots, p$

STEP 2. For $2 \leq i \leq p$,

$$\begin{aligned} r_{ii} &= \left(a_{ii} - \sum_{k=1}^{i-1} r_{ki}^2 \right)^{1/2}, \\ r_{ij} &= \left(a_{ij} - \sum_{k=1}^{i-1} r_{ki} r_{kj} \right) / r_{ii}, \quad i+1 \leq j \leq p \end{aligned}$$

(III) Spectral Decomposition

Any $n \times n$ symmetric matrix A can be written as

$$A = \Gamma D \Gamma'$$

where $\Gamma = (\mathbf{u}_1, \dots, \mathbf{u}_n)$ is $n \times n$ orthogonal matrix and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a diagonal matrix. Here, λ_i , $i = 1, \dots, n$ are eigenvalues of A , and

u_i is an eigenvector corresponding to λ_i . This decomposition is called *spectral* decomposition (or *eigen* decomposition). Further, note that we may write

$$A = \sum_{i=1}^n \lambda_i u_i u_i'$$

(IV) Singular Values Decomposition

Any $n \times p$ ($p < n$) matrix X can be written as

$$X = USV'$$

where

- (i) U is $n \times p$ orthogonal matrix consisting of eigenvectors corresponding to p largest eigenvalues among n eigenvalues of XX'
- (ii) $S = \text{diag}(s_1, \dots, s_p)$, $s_1 \geq \dots \geq s_p \geq 0$ is $p \times p$ diagonal matrix, where s_i , $i = 1, \dots, p$ is called *singular values* of X . Recall that the singular values of X is the positive square root of the eigenvalues of $X'X$ because

$$X'X = VSU'USV' = VS^2V'$$

and we see that the eigenvalues of $X'X$ are diagonal elements of S^2 by the spectral decomposition.

(iii) V is $p \times p$ orthogonal matrix consisting of the eigenvectors of $X'X$.

This decomposition is called *Singular Values Decomposition* (SVD) of $n \times p$ ($p < n$) matrix X .

A.7 Miscellanea in Matrix

A.7.1 Summing Vector and Centering Matrix

$\mathbf{1}_n = (1, 1, \dots, 1)'$ is called a *summing vector* because for an n -dimensional vector $\mathbf{x} = (x_1, \dots, x_n)'$, we have $\mathbf{1}'\mathbf{x} = \sum x_i$. Also, $\mathbf{1}_r\mathbf{1}_s'$ is $r \times s$ matrix with all the components 1, and often denoted by $J_{r \times s}$. Further, we denote $J_{n \times n}$ as J_n , and it is easy to show $J_n^2 = nJ_n$. Especially,

$$C = I - \frac{1}{n}J_n$$

is called a *centering matrix*, and we have

$$C = C' = C^2, \quad C\mathbf{1} = \mathbf{0}, \quad CJ = JC = \mathbf{0}$$

As an example for a centering matrix, consider a sample variance s^2 for $\mathbf{x} = (x_1, \dots, x_n)'$.

$$\begin{aligned} (n-1)s^2 &= \sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = \mathbf{x}'\mathbf{x} - n\left(\frac{1}{n}\mathbf{1}'\mathbf{x}\right)^2 \\ &= \mathbf{x}'\mathbf{x} - \frac{1}{n}\mathbf{x}'\mathbf{1}\mathbf{1}'\mathbf{x} = \mathbf{x}'\mathbf{x} - \frac{1}{n}\mathbf{x}'J\mathbf{x} = \mathbf{x}'\left(I - \frac{1}{n}J\right)\mathbf{x} = \mathbf{x}'C\mathbf{x} \end{aligned}$$

A.7.2 Derivatives of Matrix

(i) If x, y are n –dimensional vectors, then

$$\frac{\partial}{\partial x}(x'y) = \frac{\partial}{\partial x}(y'x) = y$$

(ii) If x is n –dimensional vector and A is $n \times n$ matrix, then

$$\begin{aligned}\frac{\partial}{\partial x}(x'A) &= A \\ \frac{\partial}{\partial x}(Ax) &= A'\end{aligned}$$

(iii) The derivative of the quadratic form of the n –dimensional vector x becomes

$$\begin{aligned}\frac{\partial}{\partial x}(x'Ax) &= Ax + A'x \\ &= 2Ax, \text{ if } A \text{ is symmetric}\end{aligned}$$

(iv) The 2nd derivative of $f(x) = f(x_1, \dots, x_n)$ w.r.t. x_i and x_j is

$$H = \left\{ \frac{\partial^2 f}{\partial x_i \partial x_j} \right\} = \frac{\partial^2 f}{\partial x \partial x'}$$

which is $n \times n$ matrix and called the *Hessian* matrix.

A.7.3 Kronecker Product

(1) definition

The *Kronecker product* or *direct product* of $A_{p \times q}$ and $B_{m \times n}$ is defined as

$$A_{p \times q} \otimes B_{m \times n} = \begin{bmatrix} a_{11}B & \cdots & a_{1q}B \\ \vdots & & \vdots \\ a_{p1}B & \cdots & a_{pq}B \end{bmatrix}$$

and which is $pm \times qn$ matrix.

(2) properties

$$(I) (A \otimes B)' = A' \otimes B'$$

$$(II) (A \otimes B)(X \otimes Y) = AX \otimes BY$$

$$(III) (A \otimes B)^{-1} = A^{-1} \otimes B^{-1}, \text{ where } A \text{ and } B \text{ are square matrices}$$

$$(IV) r(A \otimes B) = r(A)r(B)$$

$$(V) \text{tr}(A \otimes B) = \text{tr}(A)\text{tr}(B)$$

$$(VI) |A_{p \times p} \otimes B_{m \times m}| = |A|^m |B|^p$$

(VII) Eigenvalues of $A \otimes B$ is the product of the eigenvalue of A and the eigenvalue of B

A.7.4 Vec Operation

(1) definition

When we write a matrix $A_{m \times n}$ as $A = [a_1, \dots, a_n]$, where a_i is m –dimensional i th column vector, then $\text{vec}(A)$ is defined as

$$\text{vec } A = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

i.e., $\text{vec } A$ is mn –dimensional vector. For example,

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}, \quad \text{vec}(A) = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{bmatrix}$$

(2) properties

$$(I) \quad \text{vec}(ABC) = (C' \otimes A) \text{vec } B$$

$$(II) \quad \text{tr}(AB) = (\text{vec } A')' \text{vec } B$$

$$(III) \quad \text{tr}(AZ'BZC) = (\text{vec } Z')'(CA \otimes B') \text{vec } Z$$

Chapter 1. Elementary Statistical Theory

1.1 Basic Terminologies

1.1.1 Population and Sample

(i) *population* : entire set of interest

(ii) *sample* : subset of population

Ex. - population : census

- sample : sample survey

1.1.2 Variables

variable : quantitative and qualitative

- quantitative : continuous and discrete

- qualitative : nominal and ordinal

1.2 Analysis of Data

1.2.1 Tables and Graphs

(1) types of distribution

(i) symmetric

(ii) skewed to the right

(iii) skewed to the left

Figure 1.1 (p. 5)

(2) tables and graphs

(i) frequency table

(ii) histogram

(iii) boxplot

(iv) scatter diagram

1.2.2 Parameter and Statistic

(i) population - parameter

(ii) sample - statistic

1.3 Random Variables and Random Sample

random variable : a real-valued function defined on the sample space

1.3.1 Expectation and Variance of Random Variables

(1) Y : discrete r.v. with d possible values $y_1, \dots, y_d$

(i) pdf

$$f_Y(y_i) = \Pr[Y = y_i], \quad i = 1, 2, \dots, d$$

(ii) expectation

$$E[Y] = \sum_{i=1}^d y_i f_Y(y_i)$$

(2) Y : continuous r.v.

(i) pdf : $f_Y(y)$

(ii) expectation

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y) dy$$

(3) properties of expectation

(i) expectation of $g(Y)$

$$E[g(Y)] = \int_{-\infty}^{\infty} g(y) f_Y(y) dy$$

(ii) linearity property of expectation ; For constants a and b ,

$$E[a + bY] = a + bE[Y]$$

(4) variance of Y

$$\sigma_Y^2 = \text{Var}(Y) = E[(Y - E[Y])^2] = E[Y^2] - (E[Y])^2$$

(5) property of variance : For constants a and b ,

$$\text{Var}(a + bY) = b^2 \text{Var}(Y)$$

(5) standard deviation of Y : $\sigma_Y = \sqrt{\sigma_Y^2}$

(6) standardized r.v.

$$Z = \frac{Y - \mu_Y}{\sigma_Y}$$

1.3.2 Joint Probability Distribution

(1) Let $f(y_1, y_2)$ be the jpdf of Y_1 and Y_2 , then we have

$$\Pr[Y_1 \leq y_1, Y_2 \leq y_2] = \int_{-\infty}^{y_1} \int_{-\infty}^{y_2} f(y_1, y_2) dy_2 dy_1$$

(2) marginal probability distribution of Y_1

$$f_{Y_1}(y_1) = \int_{-\infty}^{\infty} f(y_1, y_2) dy_2$$

(3) conditional probability distribution of Y_1 given Y_2

$$f(y_1|Y_2 = y_2) = \frac{f(y_1, y_2)}{f_{Y_2}(y_2)}$$

(4) independence : Two r.v.'s Y_1 and Y_2 are called *independent* if

$$f(y_1, y_2) = f_{Y_1}(y_1)f_{Y_2}(y_2)$$

If Y_1 and Y_2 are independent, then

$$E[Y_1 Y_2] = E[Y_1] \cdot E[Y_2]$$

(5) Extension to n r.v.'s : If Y_i ($i = 1, 2, \dots, n$) are independent, and if

$f_{Y_i}(y)$ is the pdf of Y_i , then the jpdf is

$$f(y_1, y_2, \dots, y_n) = f_{Y_1}(y_1)f_{Y_2}(y_2) \cdots f_{Y_n}(y_n) = \prod_{i=1}^n f_{Y_i}(y_i)$$

In particular, if $Y_1, Y_2, \dots, Y_n$ are independent and have the same distribution, then they are called *random sample*.

1.3.3 Covariance and Correlation

(1) covariance between Y_1 and Y_2

$$\text{Cov}(Y_1, Y_2) = E[(Y_1 - \mu_{Y_1})(Y_2 - \mu_{Y_2})] = E[Y_1 Y_2] - \mu_{Y_1} \mu_{Y_2}$$

(2) correlation variance between Y_1 and Y_2

$$\rho = \frac{\text{Cov}(Y_1, Y_2)}{\sigma_{Y_1} \sigma_{Y_2}}$$

1.3.4 Linear Combination of Random variables

(i) $E[\sum_{i=1}^n a_i Y_i] = \sum_{i=1}^n a_i E[Y_i]$

(ii) $Var(\sum_{i=1}^n a_i Y_i) = \sum_{i=1}^n \sum_{j=1}^n a_i a_j Cov(Y_i, Y_j)$

(iii) $Var(\sum_{i=1}^n a_i Y_i) = \sum_{i=1}^n a_i^2 Var(Y_i)$ if Y_i 's are independent

(iv) $Cov(\sum_{i=1}^n a_i Y_i, \sum_{j=1}^n b_j Y_j) = \sum_{i=1}^n \sum_{j=1}^n a_i b_j Cov(Y_i, Y_j)$

(v) $Cov(\sum_{i=1}^n a_i Y_i, \sum_{j=1}^n b_j Y_j) = \sum_{i=1}^n a_i b_i Var(Y_i)$ if Y_i 's are independent

1.3.5 Discrete Probability Distribution

(1) Bernoulli trial

(i) 3 conditions for Bernoulli trials

- each trial has one of two outcomes, S (success) and F (failure)
- probability of success $p = P(S)$ should be equal in each trial
- all the trials should be independent to each other

(ii) pdf

$$f(u) = \Pr[U = u] = p^u (1 - p)^{1-u}, \quad u = 0, 1$$

(iii) mean and variance

$$E[U] = p, \quad Var[U] = p(1 - p)$$

(2) binomial distribution

(i) pdf

Y : number successes out of n Bernoulli trials, i.e., $Y = \sum_{i=1}^n U_i$

$$f(y) = \binom{n}{y} p^y (1-p)^{n-y}, \quad y = 0, 1, \dots, n$$

(iii) mean and variance

$$E[Y] = np, \quad \text{Var}(Y) = np(1-p)$$

(3) multinomial distribution

(i) pdf

$p_i, i = 1, \dots, k$: the probability for the i th category s.t. $\sum_{i=1}^k p_i = 1$

$y_i, i = 1, \dots, k$: the number of occurrences for the i th category out of n trials s.t. $\sum_{i=1}^k y_i = n$

$$f(y_1, \dots, y_{k-1}) = \frac{n!}{y_1! \dots y_k!} p_1^{y_1} \dots p_k^{y_k},$$

where $p_k = 1 - \sum_{i=1}^{k-1} p_i, \quad y_k = n - \sum_{i=1}^{k-1} y_i$

(4) Poisson distribution

(i) pdf

$$f(y) = \frac{e^{-\lambda} \lambda^y}{y!}, \quad y = 0, 1, 2, \dots$$

(ii) mean and variance

$$E[Y] = \lambda, \quad \text{Var}(Y) = \lambda$$

1.3.6 Normal Distribution

(i) pdf

$$f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right), \quad -\infty < y < \infty$$
$$Y \sim N(\mu, \sigma^2)$$

(ii) standard normal distribution

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1)$$

(iii) cdf

$$\Phi(z) = \Pr[Z \leq z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

(iv) property

If $Y_i \sim N(\mu_i, \sigma_i^2)$, $i = 1, \dots, n$, and Y_i 's are independent, then

$$\sum_{i=1}^n a_i Y_i \sim N\left(\sum_{i=1}^n a_i \mu_i, \sum_{i=1}^n a_i^2 \sigma_i^2\right)$$

(v) central limit theorem

If $Y_1, Y_2, \dots, Y_n$: r.s. from a distribution with $E(Y_1) = \mu$ and $Var(Y_1) = \sigma^2 < \infty$, then, as $n \rightarrow \infty$, we have

$$\sqrt{n}(\bar{Y} - \mu) \rightarrow N(0, \sigma^2)$$

1.3.7 Continuous R.V.'s Related with Normal Distribution

(1) χ^2 distribution

(i) def.

If $Z_1, Z_2, \dots, Z_n$ are iid $N(0,1)$, then

$$V = Z_1^2 + Z_2^2 + \cdots + Z_n^2$$

is called to have χ^2 distribution with d.f. n , and denoted by $\chi^2(n)$

(ii) mean and variance

$$E[V] = n, \quad Var(V) = 2n$$

(iii) noncentral χ^2 distribution

If $Z_i \sim N(\mu_i, 1)$ and Z_i 's are independent, then

$$V = \sum_{i=1}^n Z_i^2$$

is called to have noncentral χ^2 distribution with d.f. n and noncentrality parameter) $\lambda = \sum_{i=1}^n \mu_i^2$, and denoted by $V \sim \chi^2(n, \lambda)$. When $\lambda = 0$, then it reduces to the central χ^2 distribution.

(2) t distribution

(i) def.

If $Z \sim N(0, 1)$, $V \sim \chi^2(n)$, and Z and V are independent, then

$$t = \frac{Z}{\sqrt{V/n}}$$

is called to have a t distribution with d.f. n and denoted by $t \sim t(n)$.

(ii) property

The pdf of $t(n)$ has always heavier tails than the $N(0, 1)$ pdf, and $t(n) \rightarrow N(0, 1)$ as $n \rightarrow \infty$.

(3) F distribution

(i) def.

If $V_1 \sim \chi^2(n_1)$, $V_2 \sim \chi^2(n_2)$, and V_1 and V_2 are independent, then

$$F = \frac{V_1/n_1}{V_2/n_2}$$

is called to have F distribution with d.f's n_1 and n_2 , and denoted by $F_\alpha(n_1, n_2)$.

(ii) noncentral F distribution

If $V_1 \sim \chi^2(n_1, \lambda)$, $V_2 \sim \chi^2(n_2)$, and V_1 and V_2 are independent, then

$$F = \frac{V_1/n_1}{V_2/n_2}$$

is called to have noncentral F distribution with d.f's n_1 and n_2 , noncentral-ity parameter λ , and denoted by $F_\alpha(n_1, n_2; \lambda)$.

1.3.8 Random Vector and Multivariate Normal Distribution

(1) Random Vector

(i) def.

$\mathbf{y} = (Y_1, Y_2, \dots, Y_n)'$; n -dimensional random vector

$\boldsymbol{\mu} = E[\mathbf{y}] = (E[Y_1], E[Y_2], \dots, E[Y_n])'$: mean vector of $\mathbf{y}$

(ii) properties

When $\mathbf{A}$ is $m \times n$ matrix, $\mathbf{b}$ is m -dim vector, then

$$E[\mathbf{A}\mathbf{y} + \mathbf{b}] = \mathbf{A}E[\mathbf{y}] + \mathbf{b}$$

(iii) variance-covariance matrix of $\mathbf{y}$

$$\text{Cov}(\mathbf{y}) = E[(\mathbf{y} - \boldsymbol{\mu})(\mathbf{y} - \boldsymbol{\mu})'] = E[\mathbf{y}\mathbf{y}'] - \boldsymbol{\mu}\boldsymbol{\mu}'$$

- i th diagonal element of $\text{Cov}(\mathbf{y})$ is $\text{Var}(Y_i)$

- ij th off-diagonal element of $\text{Cov}(\mathbf{y})$ is $\text{Cov}(Y_i, Y_j)$

(iv) property of variance-covariance matrix

$$\text{Cov}(\mathbf{A}\mathbf{y} + \mathbf{b}) = \mathbf{A}\text{Cov}(\mathbf{y})\mathbf{A}'$$

(2) multivariate normal distribution

(i) def.

If $\mathbf{y} = (Y_1, Y_2, \dots, Y_n)'$ has $E[\mathbf{y}] = \boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_n)'$ and $\text{Cov}(\mathbf{y}) = \Sigma$,

and its pdf is given by

$$f(\mathbf{y}|\boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp \left[-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})'\Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu}) \right]$$

then, $\mathbf{y}$ is called to have a multivariate normal distribution with mean $\boldsymbol{\mu}$

and var-cov matrix Σ , and denoted by $\mathbf{y} \sim N_n(\boldsymbol{\mu}, \Sigma)$

(ii) properties

(a) marginal distribution

$$Y_i \sim N(\mu_i, \sigma_i^2)$$

(b) conditional distribution

Assume that $\mathbf{y} \sim N_n(\boldsymbol{\mu}, \Sigma)$, and $\mathbf{y}' = (\mathbf{y}'_1, \mathbf{y}'_2)$ such that $\mathbf{y}_1$ is r -dim.

vector. Also, we partition $\boldsymbol{\mu}, \Sigma$ in a similar way. Then, the conditional dis-

tribution of $\mathbf{y}_2$ given $\mathbf{y}_1$ is

$$y_1|y_2 \sim N_r(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(y_2 - \mu_2), \quad \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$$

(iii) bivariate case ($n = 2$)

(a) bivariate normal distribution

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix} \right),$$

where $\sigma_{12} = \rho\sigma_1\sigma_2$.

(b) conditional distribution

$$Y_1|Y_2 = y_2 \sim N(\mu_{1.2}, \sigma_{1.2}^2),$$

$$\mu_{1.2} = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(y_2 - \mu_2)$$

$$\sigma_{1.2}^2 = \sigma_1^2(1 - \rho^2)$$

1.4 Statistical Estimation and Testing

1.4.1 Types and Properties of Estimators

$\hat{\theta} \equiv \hat{\theta}(Y_1, Y_2, \dots, Y_n) : \text{estimator of } \theta$

(i) unbiasedness : $\hat{\theta}$ is called *unbiased* estimator of θ if

$$E[\hat{\theta}] = \theta.$$

Also, $E[\hat{\theta}] - \theta$ is called the *bias* of $\hat{\theta}$.

(ii) consistency : $\hat{\theta}$ is called a *consistent* estimator of θ if, $\forall \varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \Pr[|\hat{\theta} - \theta| \leq \varepsilon] = 1$$

(iii) minimum variance : $\hat{\theta}$ is called a *minimum variance* estimator of θ if,

for all the other estimators $\hat{\theta}^*$,

$$\text{Var}(\hat{\theta}) \leq \text{Var}(\hat{\theta}^*)$$

1.4.2 Least Squares Estimator

Assume that $Y_1, Y_2, \dots, Y_n$ are iid with mean θ , variance σ^2 , and

$$Y_i = \theta + \varepsilon_i, \quad i = 1, 2, \dots, n,$$

where ε_i 's are iid with mean 0, variance σ^2 . Then, $\hat{\theta}$ is called the *least squares* estimator (LSE) of θ if $\hat{\theta}$ minimizes

$$S = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (Y_i - \theta)^2$$

Now, we have

$$\sum_{i=1}^n (Y_i - \hat{\theta})^2 = \sum_{i=1}^n (Y_i - \bar{Y} + \bar{Y} - \hat{\theta})^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2 + n(\bar{Y} - \hat{\theta})^2$$

and therefore $\hat{\theta} = \bar{Y}$. On the other hand, $\hat{\theta}$ is called the *least absolute deviation* (LAD) estimator of θ if $\hat{\theta}$ minimizes

$$S = \sum_{i=1}^n |Y_i - \theta|$$

1.4.3 Maximum Likelihood Estimator

(1) Def.

Let $Y_1, Y_2, \dots, Y_n$ be iid with pdf $f(y|\theta)$, then the *likelihood function* is defined as

$$L(\theta) \equiv L(\theta|\mathbf{y}) = \prod_{i=1}^n f(y_i|\theta)$$

and $\hat{\theta}$ is called the *maximum likelihood* estimator (MLE) of θ if $\hat{\theta}$ maximizes

$L(\theta)$. Also,

$$l(\theta) \equiv \log L(\theta)$$

is called a *log – likelihood function*. The MLE can also be obtained by maximizing $l(\theta)$ since the log function is increasing.

(2) Example

Let $Y_1, Y_2, \dots, Y_n$ be r.s. from $P(\lambda)$, then

$$\begin{aligned} L(\lambda) &= \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{y_i}}{y_i!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n y_i}}{y_1! y_2! \cdots y_n!} \\ \log L(\lambda) &= -n\lambda + (\sum_{i=1}^n y_i) \log \lambda - \log(y_1! y_2! \cdots y_n!) \\ \frac{\partial \log L(\lambda)}{\partial \lambda} &= -n + \frac{1}{\lambda} \sum_{i=1}^n y_i = 0 \Rightarrow \hat{\lambda} = \bar{Y} \end{aligned}$$

(3) Properties

(i) asymptotic normality

Let θ be p –dimensional vector and $\hat{\theta}$ be its MLE, then

$$\hat{\theta} \rightarrow N_p(\theta, I^{-1}(\theta)),$$

where

$$I(\theta) = -E[\partial^2 l(\theta) / \partial \theta \partial \theta']$$

is *Fisher information matrix*.

(ii) functional invariance property

If $\hat{\theta}$ is the MLE of θ , then $g(\hat{\theta})$ is the MLE of $g(\theta)$.

1.4.4 Statistical Inference on Normal Population

Assume that $Y_1, Y_2, \dots, Y_n$ iid $N(\mu, \sigma^2)$.

(1) Point Estimation

(i) μ

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i : \text{LSE, MLE, unbiased, consistent}$$

(ii) σ^2

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 : \text{unbiased}$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2 : \text{MLE}$$

Remark : degree of freedom (d.f.) = n – number of restrictions

(2) Properties

$$(i) \bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$(ii) (n-1)s^2/\sigma^2 \sim \chi^2(n-1)$$

$$(iii) \bar{Y} \text{ and } s^2 \text{ are independent}$$

$$(iv) t = \frac{\bar{Y} - \mu}{s/\sqrt{n}} \sim t(n-1)$$

(3) Interval Estimation

$$\begin{aligned} 1 - \alpha &= \Pr[-t_{\alpha/2}(n-1) < \frac{\bar{Y} - \mu}{\text{SE}(\bar{Y})} < t_{\alpha/2}(n-1)] \\ &= \Pr[\bar{Y} - t_{\alpha/2}(n-1)\text{SE}(\bar{Y}) < \mu < \bar{Y} + t_{\alpha/2}(n-1)\text{SE}(\bar{Y})] \end{aligned}$$

Hence, $100(1 - \alpha)\%$ confidence interval of μ is

$$\bar{Y} - t_{\alpha/2}(n-1)SE(\bar{Y}) < \mu < \bar{Y} + t_{\alpha/2}(n-1)SE(\bar{Y})$$

(4) Testing

null hypothesis	test statistic	alternative hypothesis	rejection region
$H_0 : \mu = \mu_0$	$t = \frac{\bar{Y} - \mu_0}{SE(\bar{Y})}$	$H_1 : \mu < \mu_0$	$t < -t_{\alpha}(n-1)$
		$H_1 : \mu > \mu_0$	$t > t_{\alpha}(n-1)$
		$H_1 : \mu \neq \mu_0$	$ t > t_{\alpha/2}(n-1)$

1.5 Taylor Approximation

1.5.1 Taylor Series Expansion

(1) $f : R \rightarrow R$ case

If $f : R \rightarrow R$ is k times differentiable at x_0 , and $|x - x_0| \rightarrow 0$, then

$$f(x) = \sum_{j=0}^k \frac{1}{j!} f^{(j)}(x_0)(x - x_0)^j + o(|x - x_0|^k),$$

where $f^{(j)}(x) = d^j f(x) / dx^j$. Further, if f is $k+1$ times differentiable and

$$|f^{(k+1)}(x)| \leq M < \infty$$

and we can write

$$f(x) = \sum_{j=0}^k \frac{1}{j!} f^{(j)}(x_0)(x - x_0)^j + \frac{1}{(k+1)!} f^{(k+1)}(\xi)(x - x_0)^{k+1},$$

where ξ is between x and x_0 .

(2) $f : R^d \rightarrow R$ case

If $f : R^d \rightarrow R$ is k times differentiable at $x_0 \in R^d$ and $\|x - x_0\| \rightarrow 0$, then

$$f(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)'(\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)'H(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) + \cdots + o(\|\mathbf{x} - \mathbf{x}_0\|^k),$$

where $\nabla f(\mathbf{x}) = \partial f(\mathbf{x})/\partial \mathbf{x}$ and $H(\mathbf{x})$ is the Hessian matrix.

(3) $f : R^d \rightarrow R^p$ case

If $f : R^d \rightarrow R^p$ is differentiable at $\mathbf{x}_0 \in R^d$ and $\|\mathbf{x} - \mathbf{x}_0\| \rightarrow 0$, then

$$f(\mathbf{x}) = f(\mathbf{x}_0) + Df(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) + o(\|\mathbf{x} - \mathbf{x}_0\|),$$

where $f(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_p(\mathbf{x}))'$ and $Df(\mathbf{x})$ is a $p \times d$ matrix with $\partial f_i(\mathbf{x})/\partial x_j$ as its (i, j) th component.

1.5.2 Delta Method

(1) univariate case

Let X be a r.v. with mean μ , variance σ^2 . We are interested in computing the mean and variance of $f(X)$ for some function f which is differentiable.

In many cases, we cannot compute the mean and variance of $f(X)$ exactly.

In this case, we are only to obtain the approximate mean and variance of $f(X)$ by using the Taylor expansion up to the linear term, i.e.,

$$f(X) \simeq f(\mu) + (X - \mu)f'(\mu)$$

Now, we can easily show that

$$E[f(X)] \simeq f(\mu)$$

$$\text{Var}[f(X)] \simeq \sigma^2 f'(\mu)^2$$

which is called *delta method* or the *useful limit theorem*.

(2) multivariate case

We extend the delta method to the d -dimensional random vector. Let $\mathbf{X}$ be d -dimensional random vector with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$.

Assume that $\mathbf{f} : \mathbb{R}^d \rightarrow \mathbb{R}^p$ is differentiable at $\boldsymbol{\mu}$, then we have

$$E[\mathbf{f}(\mathbf{X})] \simeq \mathbf{f}(\boldsymbol{\mu})$$

$$\text{Cov}[\mathbf{f}(\mathbf{X})] \simeq D\mathbf{f}(\boldsymbol{\mu})'\boldsymbol{\Sigma}D\mathbf{f}(\boldsymbol{\mu})$$

(3) Example

Let $Y_1, Y_2, \dots, Y_n$ be iid with mean μ , variance σ^2 . Then, by CLT, as $n \rightarrow \infty$,

$$\sqrt{n}(\bar{Y} - \mu) \longrightarrow N(0, \sigma^2)$$

Then, by the delta method, we have

$$\sqrt{n}(\bar{Y}^2 - \mu^2) \longrightarrow N(0, 4\mu^2\sigma^2)$$

1.5.3 Newton-Raphson Method

Recall that MLE can be obtained by maximizing $L(\theta)$ or $l(\theta) = \log L(\theta)$, and in many cases $\partial l(\theta)/\partial \theta = 0$ (called *score function*) gives the explicit solution. But, it is quite often that the score function does not give ana-

lytic (closed form) solution. In this case, we use an iteration method, called *Newton Raphson* method.

Let $\boldsymbol{\theta}$ be p -dimensional vector, and let

$$\nabla l(\boldsymbol{\theta}) = \partial l(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}$$

$$H(\boldsymbol{\theta}) = \partial^2 l(\boldsymbol{\theta}) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'$$

Since the MLE satisfies $\mathbf{0} = \nabla l(\boldsymbol{\theta})$, and the score vector does not have analytic solution, we expand $\nabla l(\boldsymbol{\theta})$ about the initial vector $\hat{\boldsymbol{\theta}}^{(0)}$ by the Taylor expansion,

$$\mathbf{0} = \nabla l(\boldsymbol{\theta}) \simeq \nabla l(\hat{\boldsymbol{\theta}}^{(0)}) + \mathbf{H}(\hat{\boldsymbol{\theta}}^{(0)})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}^{(0)}),$$

and let $\hat{\boldsymbol{\theta}}^{(1)}$ be the solution of $\boldsymbol{\theta}$ in the above equation, then

$$\hat{\boldsymbol{\theta}}^{(1)} = \hat{\boldsymbol{\theta}}^{(0)} - H(\hat{\boldsymbol{\theta}}^{(0)})^{-1} \nabla l(\hat{\boldsymbol{\theta}}^{(0)})$$

and we iterate until convergence, i.e.,

$$\hat{\boldsymbol{\theta}}^{(t+1)} = \hat{\boldsymbol{\theta}}^{(t)} - H(\hat{\boldsymbol{\theta}}^{(t)})^{-1} \nabla l(\hat{\boldsymbol{\theta}}^{(t)}), \quad t = 0, 1, 2, \dots$$

Remark 1. If we use the Fisher information matrix $-E[H(\boldsymbol{\theta})]$ instead of $-H(\boldsymbol{\theta})$, it is called the *Fisher's scoring* method.

Remark 2. For a scalar θ , we have

$$H(\hat{\theta}^{(t)}) = -\nabla l(\hat{\theta}^{(t)}) / (\hat{\theta}^{(t+1)} - \hat{\theta}^{(t)})$$

Fig. 1.4 (P. 33)

1.6 Multiple Testing

1.6.1 Examples of Multiple Testing

(1) oneway classification

Consider oneway classification model with 3 treatments

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad i = 1, 2, 3, \quad j = 1, 2, \dots, n$$

and consider the following hypotheses

$$H_0 : \mu_1 = \mu_2 = \mu_3 \quad vs \quad H_1 : \text{not } H_0$$

which is equivalent to

$$H_{01} : \mu_1 = \mu_2, \quad H_{02} : \mu_1 = \mu_3, \quad H_{03} : \mu_2 = \mu_3$$

Simultaneous tests for testing more than one hypothesis is called *multiple testing*

which is equivalent to computing *simultaneous confidence interval* for

$$\mu_1 - \mu_2, \quad \mu_1 - \mu_3, \quad \mu_2 - \mu_3,$$

and it also called *multiple comparison*.

(2) microarray data

Let n_1, n_2 be number of lung cancer patients and normal people, respectively, where $n = n_1 + n_2$. Also, let the microarray data is given by the $p \times n$ matrix, where p is the number of genes. Often n is tens or hundreds and p is thousands or tens of thousands, and this type of data is called "small n , large p " or *high – dimensional data*. Let μ_{1i} denote mean of gene

expression of lung cancer patients and μ_{2i} denote mean of gene expression of normal people. Hence, we are interested in testing

$$H_{0i} : \mu_{1i} = \mu_{2i}, \quad i = 1, 2, \dots, p$$

1.6.2 Type I Error

Consider m multiple testing problem

$$H_{0i} : \mu_{1i} = \mu_{2i}, \quad i = 1, 2, \dots, m$$

and denote E_i as an event of rejecting H_{0i} even though H_{0i} is true. When the level of significance is α , and if a test satisfies $P(E_i) \leq \alpha$, then the test is called to be *controlled* at α . Now, $P(E_i)$ is the probability of type I error for the i th test. In multiple testing the probability of type I error is rejecting at least one hypothesis even though H_{0i} , $i = 1, 2, \dots, m$ is true, i.e., the probability of type I error in m multiple testing is

$$P(\cup E_i) = 1 - P(\cap E_i^c) = 1 - \prod_{i=1}^m P(E_i^c) = 1 - (1 - \alpha)^m$$

For example, when $\alpha = .05$,

$$m = 2 \quad : \quad 1 - (1 - 0.05)^2 = 0.0975$$

$$m = 3 \quad : \quad 1 - (1 - 0.05)^3 = 0.1426$$

$$m = 100 \quad : \quad 1 - (1 - 0.05)^{100} = 0.9941$$

1.6.3 Family-Wise Error Rate

fact test result	do not reject H_0	reject H_0	total
H_0 true	U	V	m_0
H_1 true	T	S	m_1
total	$m - R$	R	m

FWER (family-wise error rate) controls α s.t.

$$P(V \geq 1 | m_0 > 0) \leq \alpha$$

(1) Bonferroni's method

Bonferroni method is based on the *Bonferroni inequality*;

$$P(V \geq 1) = P(\cup E_i) \leq \sum P(E_i) \leq \alpha$$

Therefore, to control $P(V \geq 1) \leq \alpha$, we need to control $P(E_i) \leq \alpha^* \equiv \alpha/m$.

For example, when $\alpha = .05$ and 10 multiple hypotheses are given, then we have set the level of significance at $.05/10 = .005$. Let p_i be the p -value for the i hypothesis, then we have to change the p -value to $p_i^* = mp_i$ when we test m multiple hypotheses.

Therefore, Bonferroni method rejects H_{0i} if $P(E_i) \leq \alpha/m$, or equivalently if $mp_i \leq \alpha$, i.e., Bonferroni method is too conservative, and consequently, it is not a powerful test.

(2) Sidak method

Sidak method is based on the following equation;

$$P(\cup E_i) = 1 - (1 - \alpha)^m.$$

Hence, $1 - (1 - \alpha^*)^m \leq \alpha$ is equivalent to $\alpha^* \leq 1 - (1 - \alpha)^{1/m}$, instead of α , we have to use $\alpha^* = 1 - (1 - \alpha)^{1/m}$. In the sense of p -values, it becomes $p_i^* = 1 - (1 - p_i)^m$. Even though the Sidak method is more powerful than the Benferroni method, it is still conservative, too.

1.6.4 False Discovery Rate

The classical FWER, based on Fisher - Neyman - Pearson, is focused on maximizing the power subject to the type I error, however, it is not an adequate method to the multiple tests in high-dimensional data. This problem was solved by Benjamini and Hochberg (1995) by the *FDR* (false discovery rate). Instead of controlling the type I error, FDR controls

$$E(V/R) \leq q,$$

where q , $0 < q < 1$, is a given constant. For example, if $\alpha = q = .05$, $m = 1,000$ and we reject 20 hypotheses out of 1,000. For FWER, the probability of the falsely rejected null hypotheses are at least one is less than $\alpha = .05$. For FDR, the expected value of the falsely rejected null hypotheses is at most one.

Theorem 1.2 Let $p_1 \leq p_2 \leq \dots \leq p_m$ be the p -values of m hypotheses, and q is given, and if

$$j = \max_{1 \leq i \leq m} \left\{ i \mid p_i \leq \frac{i}{m} q \right\}$$

then, FDR rejects hypotheses corresponding to $p_1 \leq p_2 \leq \dots \leq p_j$.

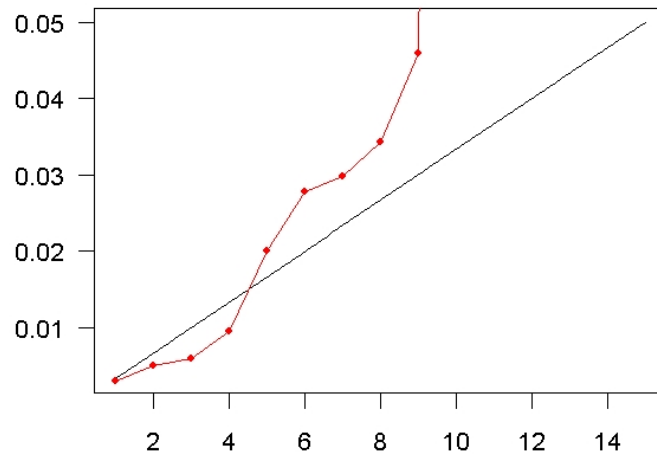
For example, assume that $m = 15$, $\alpha = q = 0.05$, and $p_1 \leq p_2 \leq \dots \leq p_{15}$

are given as follows;

0.0030	0.0050	0.0060	0.0095	0.0201	0.0278	0.0298	0.0344
0.0459	0.3240	0.4262	0.5719	0.6528	0.7590	1.0000	

Since $0.05/15 = 0.0033$, Bonferroni method rejects only one hypothesis,

but FDR rejects 4 hypotheses. (See Figure 1.5 in p. 38)



(Figure 1.5) FDR

Chapter 2. Simple Linear Regression Model

2.1 Basic Concepts of Regression Analysis

(1) types of models

(i) deterministic (mathematical) model

- model with no errors

- $E = mc^2$, $Y = \frac{1}{2}gt^2$, ...

(ii) statistical model

- model with errors

- $Y = f(X) + \epsilon$

(2) regression models

- relationship between the response Y and covariates X

- *regression* : Francis Galton (1885) "regression towards mediocrity"

(3) goals of regression analysis

- appropriate model building

- estimation of parameters in the regression model

- prediction based on the fitted regression model

2.2 Simple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, 2, \dots, n$$

ε_i : error terms, iid with $E(\varepsilon_i) = 0$ and $Var(\varepsilon_i) = \sigma^2$.

Y_i : independent with $E(Y_i) = \beta_0 + \beta_1 X_i$ and $Var(Y_i) = \sigma^2$

β_0, β_1 : regression coefficients to be estimated

2.3 Estimation of Regression Coefficients

2.3.1 Least squares Estimation

(1) LSE

(i) Def. $\hat{\beta}_0, \hat{\beta}_1$ are called the *LSE* (least squares estimator) of β_0 and β_1 , respectively, if they minimize

$$S = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2.$$

By the partial derivative of S w.r.t. β_0 and β_1 , respectively,

$$\begin{aligned} \frac{\partial S}{\partial \beta_0} &= (-2) \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i) = 0 \\ \frac{\partial S}{\partial \beta_1} &= (-2) \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i) X_i = 0 \end{aligned}$$

Hence, we obtain

$$\begin{aligned} n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n X_i &= \sum_{i=1}^n Y_i \\ \hat{\beta}_0 \sum_{i=1}^n X_i + \hat{\beta}_1 \sum_{i=1}^n X_i^2 &= \sum_{i=1}^n X_i Y_i \end{aligned}$$

which are called *normal equation*. Finally, we have

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}, \quad \hat{\beta}_1 = \frac{S_{XY}}{S_{XX}},$$

where $S_{XY} = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$, $S_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2$.

(ii) terminologies

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i : \text{fitted value at } X_i$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x : \text{fitted regression line}$$

$$e_i = Y_i - \hat{Y}_i : \text{the } i\text{th residual}$$

See Fig. 2.4 (p. 50)

(2) LAD

Estimators minimizing

$$\sum_{i=1}^n |\varepsilon_i| = \sum_{i=1}^n |Y_i - \beta_0 - \beta_1 X_i|$$

are called *LAD* (least absolute deviation) estimator of β_0 and β_1 .

Ex. 2.2 (p.52)

2.3.2 Properties of LSE

For the time being, we assume that the covariate X is constant, not random variable. Therefore, random variables in the simple linear regression model are the response Y and error terms ϵ .

(1) expectation

First, note that

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})Y_i - \bar{Y} \sum_{i=1}^n (X_i - \bar{X})}{S_{XX}} = \sum_{i=1}^n w_i Y_i,$$

where $w_i = \frac{(X_i - \bar{X})}{S_{XX}}$. Hence,

$$E[\hat{\beta}_1] = E\left[\sum_{i=1}^n w_i Y_i\right] = \sum_{i=1}^n w_i (\beta_0 + \beta_1 X_i) = \beta_0 \sum_{i=1}^n w_i + \beta_1 \sum_{i=1}^n w_i X_i = \beta_1$$

because $\sum_{i=1}^n w_i = 0$, $\sum_{i=1}^n w_i X_i = 1$. Also,

$$E[\hat{\beta}_0] = E[\bar{Y} - \hat{\beta}_1 \bar{X}] = \frac{1}{n} E\left[\sum_{i=1}^n Y_i\right] - E[\hat{\beta}_1] \bar{X} = \frac{1}{n} \sum_{i=1}^n (\beta_0 + \beta_1 X_i) - \beta_1 \bar{X} = \beta_0$$

Conclusively, $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased estimators of β_0 and β_1 , respectively.

(2) variance

$$\text{Var}(\hat{\beta}_1) = \text{Var}\left[\sum_{i=1}^n w_i Y_i\right] = \sigma^2 \sum_{i=1}^n w_i^2 = \frac{\sigma^2}{S_{XX}}$$

Since

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = \frac{1}{n} \sum_{i=1}^n Y_i - \left(\sum_{i=1}^n w_i Y_i\right) \bar{X} = \sum_{i=1}^n \left(\frac{1}{n} - w_i \bar{X}\right) Y_i = \sum_{i=1}^n w_i^* Y_i,$$

where $w_i^* = \frac{1}{n} - w_i \bar{X}$, we have

$$\text{Var}(\hat{\beta}_0) = \text{Var}\left(\sum_{i=1}^n w_i^* Y_i\right) = \sigma^2 \sum_{i=1}^n (w_i^*)^2 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{XX}}\right)$$

Theorem 2.1 (Gauss-Markov Theorem)

$\hat{\beta}_0, \hat{\beta}_1$, LSE of β_0, β_1 , has the minimum variances among all the unbiased and linear (in Y_i 's) estimators, i.e., $\hat{\beta}_0, \hat{\beta}_1$ are BLUE (best linear unbiased estimator) of β_0, β_1 .

(Proof)

Let $\hat{\beta}_1^*$ be any other unbiased and linear estimator of β_1 , i.e.,

$$\hat{\beta}_1^* = \sum_{i=1}^n c_i Y_i,$$

then there exist constants d_i such that $c_i = w_i + d_i$. Therefore,

$$\hat{\beta}_1^* = \sum_{i=1}^n (w_i + d_i) Y_i$$

and

$$\begin{aligned} E[\hat{\beta}_1^*] &= \sum_{i=1}^n (w_i + d_i) E[Y_i] \\ &= \sum_{i=1}^n (w_i + d_i) (\beta_0 + \beta_1 X_i) \\ &= \beta_0 \sum_{i=1}^n (w_i + d_i) + \beta_1 \sum_{i=1}^n (w_i + d_i) X_i \end{aligned}$$

Now, $\sum_{i=1}^n w_i = 0$ and $\sum_{i=1}^n w_i X_i = 1$ give

$$\sum_{i=1}^n d_i = 0, \quad \sum_{i=1}^n d_i X_i = 0$$

Also, the variance of $\hat{\beta}_1^*$ is

$$\begin{aligned}
Var(\hat{\beta}_1^*) &= \sigma^2 \sum_{i=1}^n c_i^2 \\
&= \sigma^2 \sum_{i=1}^n (w_i + d_i)^2 \\
&= \sigma^2 \left(\sum_{i=1}^n w_i^2 + \sum_{i=1}^n d_i^2 + 2 \sum_{i=1}^n w_i d_i \right) \\
&= Var(\hat{\beta}_1) + \sigma^2 \sum_{i=1}^n d_i^2 + 2\sigma^2 \sum_{i=1}^n w_i d_i,
\end{aligned}$$

where

$$\sum_{i=1}^n w_i d_i = \frac{\sum_{i=1}^n (X_i - \bar{X}) d_i}{S_{XX}} = \frac{\sum_{i=1}^n d_i X_i - \bar{X} \sum_{i=1}^n d_i}{S_{XX}} = 0$$

Hence, we must have

$$Var(\hat{\beta}_1) < Var(\hat{\beta}_1^*)$$

which completes the proof.

(3) other properties

Since

$$Cov(\bar{Y}, \hat{\beta}_1) = Cov\left(\frac{1}{n} \sum_{i=1}^n Y_i, \sum_{i=1}^n w_i Y_i\right) = \sum_{i=1}^n \frac{1}{n} w_i Var(Y_i) = \frac{\sigma^2}{n} \sum_{i=1}^n w_i = 0$$

we have

$$Var(\hat{\beta}_0) = Var(\bar{Y} - \hat{\beta}_1 \bar{X}) = Var(\bar{Y}) + \bar{X}^2 Var(\hat{\beta}_1) = \frac{\sigma^2}{n} + \frac{\bar{X}^2}{S_{XX}} \sigma^2$$

Useful facts

- (1) $\sum_{i=1}^n e_i = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$: sum of residuals is 0.
- (2) $\sum_{i=1}^n X_i e_i = 0$: sum of weighted average of X_i and residuals is 0.
- (3) $\sum_{i=1}^n \hat{Y}_i e_i = 0$: sum of weighted average of $\hat{Y}_i$ and residuals is 0 .
- (4) The fitted line always go through $(\bar{X}, \bar{Y})$.

2.3.3 Estimation of σ^2

$$s^2 = \frac{1}{n-2} \sum_{i=1}^n e_i^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

2.3.4 Maximum Likelihood Estimation

Assume that ε_i 's are iid $N(0, \sigma^2)$. Then, the likelihood function is

$$\begin{aligned} L &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} \varepsilon_i^2\right) \\ &= \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2\right] \end{aligned}$$

For fixed σ^2 , maximizing L w.r.t. β_0 and β_1 is equivalent to minimizing $\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$. Hence, MLEs of β_0 and β_1 are equal to the LSEs of β_0 and β_1 . Now, the MLE of σ^2 is given by

$$\frac{\partial \log L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

so that

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Note that $\hat{\sigma}^2 = \frac{n-2}{n} s^2$.

Remark. We showed that MLEs of β_0 and β_1 are equal to the LSEs of β_0 and β_1 when the error distribution is normal. On the other hand, MLEs of β_0 and β_1 are equal to the LADs of β_0 and β_1 when the error distribution is the double exponential distribution (Laplace distribution), i.e.,

$$f(\varepsilon) = \frac{1}{2\sigma} \exp(-|\varepsilon|/\sigma)$$

2.4 Goodness-of-Fit of the Regression Line

As an overall goodness-of-fit measures, the standard error of estimate

$$s = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

can be used, however, it is not scale-invariant. Instead we often use the *coefficient of determination* R^2 , and will derive in the following;

From the following identity

$$Y_i - \bar{Y} = (Y_i - \hat{Y}_i) + (\hat{Y}_i - \bar{Y}), \quad i = 1, 2, \dots, n$$

we have

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2, \quad (1)$$

where

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 : \text{SST (total sum of squares), d.f.} = (n - 1)$$

$$\sum_{i=1}^n (Y_i - \hat{Y}_i)^2 : \text{SSE (error sum of squares), d.f.} = (n - 2)$$

$$\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 : \text{SSR (regression sum of squares), d.f.} = 1$$

Remark. $\text{SSR} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \hat{\beta}_1^2 \sum_{i=1}^n (X_i - \bar{X})^2$.

Def. $R^2 = \frac{\text{SSR}}{\text{SST}}$: *coefficient of determination*

2.5 Analysis of Variance

source	SS	d.f.	MS	F-ratio
regression	SSR	1	MSR=SSR/1	$F_0 = \text{MSR} / \text{MSE}$
error	SSE	$(n - 2)$	MSE=SSE / $(n - 2)$	
total	SST	$(n - 1)$		

hypotheses : $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$

test statistic : $F_0 = \frac{(\text{SSR} / \sigma^2) / 1}{(\text{SSE} / \sigma^2) / (n - 2)} = \frac{\text{MSR}}{\text{MSE}}$

Reject H_0 if $F_0 > F_\alpha(1, n - 2)$

2.6 Statistical Inference

2.6.1 Inference on β_1

(1) distribution of $\hat{\beta}_1$

Since $Var(\hat{\beta}_1) = \sigma^2 / S_{XX}$, we have

$$\frac{\hat{\beta}_1 - \beta_1}{\sigma / \sqrt{S_{XX}}} \sim N(0, 1)$$

and $SE(\hat{\beta}_1) = s / \sqrt{S_{XX}}$. Therefore,

$$t = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1}{s / \sqrt{S_{XX}}} \sim t(n - 2)$$

because

$$t = \frac{\frac{\hat{\beta}_1 - \beta_1}{\sigma / \sqrt{S_{XX}}}}{\sqrt{\frac{SSE}{\sigma^2(n - 2)}}}$$

(2) C.I. of β_1

Since

$$\Pr[-t_{\alpha/2}(n - 2) < \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} < t_{\alpha/2}(n - 2)] = 1 - \alpha$$

100(1 - α)% C.I. for β_1 is

$$\hat{\beta}_1 - t_{\alpha/2}(n - 2)SE(\hat{\beta}_1) < \beta_1 < \hat{\beta}_1 + t_{\alpha/2}(n - 2)SE(\hat{\beta}_1)$$

(3) Test for β_1

hypothesis : $H_0 : \beta_1 = \beta_{1,0}$

test statistic : $t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} \sim t(n-2)$ under H_0

null hypothesis	test statistic	alternative hypothesis	rejection region
$H_0 : \beta_1 = \beta_{1,0}$	$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)}$	$H_1 : \beta_1 < \beta_{1,0}$	$t < -t_{\alpha}(n-2)$
		$H_1 : \beta_1 > \beta_{1,0}$	$t > t_{\alpha}(n-2)$
		$H_1 : \beta_1 \neq \beta_{1,0}$	$ t > t_{\alpha/2}(n-2)$

2.6.2 Inference on β_0

(1) distribution of $\hat{\beta}_0$

Since $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, which is a linear combination of $Y_1, Y_2, \dots, Y_n$, so that

$$\hat{\beta}_0 \sim N(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{XX}} \right)).$$

Now, σ^2 is unknown, we replace it by s^2 , and the standard error of $\hat{\beta}_0$ is

$$SE(\hat{\beta}_0) = s \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{S_{XX}}}.$$

Therefore,

$$t = \frac{\hat{\beta}_0 - \beta_0}{SE(\hat{\beta}_0)} \sim t(n-2)$$

(2) C.I. of $\beta_0 : \hat{\beta}_0 \pm t_{\alpha/2}(n-2)SE(\hat{\beta}_0)$

(3) Test for β_0

hypothesis : $H_0 : \beta_0 = \beta_{0,0}$

test statistic : $t = \frac{\hat{\beta}_0 - \beta_{0,0}}{\text{SE}(\hat{\beta}_0)} \sim t(n-2)$ under H_0

null hypothesis	test statistic	alternative hypothesis	rejection region
$H_0 : \beta_0 = \beta_{0,0}$	$t = \frac{\hat{\beta}_0 - \beta_{0,0}}{\text{SE}(\hat{\beta}_0)}$	$H_1 : \beta_0 < \beta_{0,0}$	$t < -t_{\alpha}(n-2)$
		$H_1 : \beta_0 > \beta_{0,0}$	$t > t_{\alpha}(n-2)$
		$H_1 : \beta_0 \neq \beta_{0,0}$	$ t > t_{\alpha/2}(n-2)$

2.6.3 Inference on the Response

(1) Inference on $E(Y) = \beta_0 + \beta_1 x$

As a point estimator of $E(Y) = \beta_0 + \beta_1 x$, it is reasonable to use $\hat{Y}(x) =$

$\hat{\beta}_0 + \hat{\beta}_1 x$ because $E[\hat{Y}(x)] = \beta_0 + \beta_1 x$. Further,

$$\text{Var}(\hat{Y}(x)) = \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x) = \text{Var}(\bar{Y} + \hat{\beta}_1(x - \bar{X}))$$

and because $\bar{Y}$ and $\hat{\beta}_1$ are independent (see 2.3.2),

$$\text{Var}(\hat{Y}(x)) = \text{Var}(\bar{Y}) + (x - \bar{X})^2 \text{Var}(\hat{\beta}_1) = \sigma^2 \left[\frac{1}{n} + \frac{(x - \bar{X})^2}{S_{XX}} \right]$$

Also,

$$\text{SE}(\hat{Y}(x)) = s \sqrt{\frac{1}{n} + \frac{(x - \bar{X})^2}{S_{XX}}}$$

Hence, for example, $100(1 - \alpha)\%$ C.I. for $E(Y) = \beta_0 + \beta_1 x$ is

$$(\hat{\beta}_0 + \hat{\beta}_1 x) \pm t_{\alpha/2}(n-2) \text{SE}(\hat{Y}(x))$$

Remark. Set of C.I.'s for all x are called *confidence band*. (See Fig. 2.8 (p.74))

(2) Inference on a Particular Response $Y(x) = \beta_0 + \beta_1 x$

Since $Y(x) \sim N(\beta_0 + \beta_1 x, \sigma^2)$, it is reasonable to use $\hat{Y}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$ as

an estimator for $Y(x)$, because $Y(x) - \hat{Y}(x)$ has its mean

$$E[Y(x) - \hat{Y}(x)] = (\beta_0 + \beta_1 x) - (\beta_0 + \beta_1 x) = 0$$

and its variance

$$Var(Y(x) - \hat{Y}(x)) = Var(Y(x)) + Var(\hat{Y}(x)) = \sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x - \bar{X})^2}{S_{XX}} \right]$$

Therefore,

$$\frac{Y(x) - \hat{Y}(x)}{\sigma \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{X})^2}{S_{XX}}}} \sim N(0, 1)$$

and

$$\frac{Y(x) - \hat{Y}(x)}{s \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{X})^2}{S_{XX}}}} \sim t(n - 2)$$

The $100(1 - \alpha)\%$ *prediction interval* for $Y(x)$ is

$$\hat{Y}(x) \pm t_{\alpha/2}(n - 2) \cdot s \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{X})^2}{S_{XX}}}$$

Ex. 2.8 (p.73)

2.7 Residual Analysis

(1) Basic Assumptions in the Linear Regression

- (i) linearity : $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$
- (ii) mean of error terms : $E[\varepsilon_i] = 0, i = 1, 2, \dots, n$ (systematic deviation)
- (iii) equal variance : $Var(\varepsilon_i) = \sigma^2, i = 1, 2, \dots, n$ (homoscedasticity)
cf. unequal variance (heteroscedasticity)
- (iv) independence : $Y_1, Y_2, \dots, Y_n$ are independent
- (v) normality : ε_i 's are normally distributed
- (vi) X is not random but constant

(2) Residual Analysis

- (i) normal probability plot : normality of residuals
- (ii) $Q - Q$ (quantile-to-quantile) plot : quantiles of residuals vs quantiles of assumed distribution

Ex. 2.9 (p.77), Ex. 2.10 (p.79)

2.8 Straight Line through the Origin

(1) Straight Line through the Origin

model : $Y_i = \beta_1 X_i + \varepsilon_i, i = 1, 2, \dots, n$

LSE : minimize $S = \sum_{i=1}^n (Y_i - \beta_1 X_i)^2 \Rightarrow \hat{\beta}_1 = \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2}$

(2) Properties

- (i) $E[\hat{\beta}_1] = \beta_1$, $Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n X_i^2}$
- (ii) estimator of σ^2 : $s^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$
- (iii) $SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum_{i=1}^n X_i^2}}$
- (iv) $t = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \sim t(n-1)$
- (v) $100 \times (1 - \alpha)\%$ C.I. for β_1 : $\hat{\beta}_1 \pm t_{\alpha/2}(n-1)SE(\hat{\beta}_1)$

(3) ANOVA table

- (i) decomposition of sum of squares: $\sum_{i=1}^n Y_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n \hat{Y}_i^2$

(ii) ANOVA table

source	s.s	d.f.	mean squares	F-ratio
regression	SSR	1	MSR=SSR/1	$F_0 = \text{MSR}/\text{MSE}$
error	SSE	$(n-1)$	$\text{MSE} = \text{SSE}/(n-1)$	
total	SST	n		

- (iii) $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0 \Rightarrow$ Reject H_0 if $F_0 > F_{\alpha}(1, n-1)$

- (iv) $R^2 = \text{SSR} / \text{SST} = \hat{\beta}_1^2 \sum_{i=1}^n X_i^2 / \sum_{i=1}^n Y_i^2$

Ex. 2.11 (p.84)

2.9 Correlation Analysis

Assume that $(X, Y)'$ follows the bivariate normal distribution with mean

$(\mu_X, \mu_Y)'$ and variance-covariance matrix

$$\begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix}$$

then the conditional distribution of Y given X follows a univariate normal distribution with mean

$$\mu_{Y|X} = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (X - \mu_X)$$

and variance

$$\sigma_{Y|X}^2 = \sigma_Y^2(1 - \rho^2).$$

Therefore, we see that

$$E[Y|X] = \beta_0 + \beta_1 X, \quad (2)$$

where $\beta_0 = \mu_Y - \rho \frac{\sigma_Y}{\sigma_X} \mu_X$, $\beta_1 = \rho \frac{\sigma_Y}{\sigma_X}$.

2.9.1 Estimation of the Correlation Coefficient

Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be iid copies of the bivariate normal distribution, then the sample correlation coefficient is

$$r = \frac{S_{XY}}{\sqrt{S_{XX}}\sqrt{S_{YY}}} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}}.$$

Also, we have

$$r = \hat{\beta}_1 \sqrt{\frac{S_{XX}}{S_{YY}}}.$$

Remark $r^2 = R^2$

2.9.2 Test of the Correlation Coefficient

(1) $H_0 : \rho = 0$

Under H_0 , we have

$$t = \sqrt{n-2} \frac{r}{\sqrt{1-r^2}} \sim t(n-2)$$

null hypo.	test statistic	alt. hypo.	rej. reg.
$H_0 : \rho = 0$	$t = \sqrt{n-2} \frac{r}{\sqrt{1-r^2}}$	$H_1 : \rho < 0$	$t < -t_\alpha(n-2)$
		$H_1 : \rho > 0$	$t > t_\alpha(n-2)$
		$H_1 : \rho \neq 0$	$ t > t_{\alpha/2}(n-2)$

(2) $H_0 : \rho = \rho_0$

Under H_0 , we have, as $n \rightarrow \infty$

$$\frac{1}{2} \ln \frac{1+r}{1-r} \rightarrow N\left(\frac{1}{2} \ln \frac{1+\rho}{1-\rho}, \frac{1}{n-3}\right),$$

i.e.,

$$Z = \frac{\frac{1}{2} \ln \frac{1+r}{1-r} - \frac{1}{2} \ln \frac{1+\rho}{1-\rho}}{1/\sqrt{n-3}} = \frac{1}{2} \sqrt{n-3} \ln \left[\frac{(1+r)(1-\rho)}{(1-r)(1+\rho)} \right] \rightarrow N(0,1)$$

null hypo.	test statistic	alt. hypo.	rej. reg.
$H_0 : \rho = \rho_0$	$Z = \frac{\sqrt{n-3}}{2} \ln \left[\frac{(1+r)(1-\rho_0)}{(1-r)(1+\rho_0)} \right]$	$H_1 : \rho < \rho_0$	$Z < -z_\alpha$
		$H_1 : \rho > \rho_0$	$Z > z_\alpha$
		$H_1 : \rho \neq \rho_0$	$ Z > z_{\alpha/2}$

Chapter 3. Multiple Linear Regression Model

3.1 Multiple Linear Regression Model

3.1.1 Population Multiple Linear Regression Model

Assume that

$$E[Y] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_{p-1} X_{p-1}$$

then, we have the following population multiple linear regression model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i, \quad i = 1, 2, \cdots, n$$

3.1.2 Linear and Nonlinear Models

Consider a regression model

$$Y = f(x, \beta) + \varepsilon.$$

There are two kinds of regression models, *linear* and *nonlinear*. If $\partial f / \partial \beta_j, \forall j$ does not contain regression coefficients, then the model is called linear.

Otherwise, it is called nonlinear. For example,

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon; \quad Y = \beta_0 + \beta_1 \log X_1 + \beta_2 \log X_2 + \varepsilon$$

are linear, but

$$Y = \beta_0 + \beta_1 e^{-\beta_2 x} + \varepsilon$$

is nonlinear.

3.1.3 Regression Model Using Matrix

Let $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_{p-1})'$ and $\mathbf{x}_i = (1, X_{i1}, \dots, X_{i,p-1})'$, then the multiple linear regression model can be written as

$$Y_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

If we use the matrix notations, we may write

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

$$\mathbf{y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} \mathbf{x}_1' \\ \mathbf{x}_2' \\ \vdots \\ \mathbf{x}_n' \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1,p-1} \\ 1 & X_{21} & X_{22} & \cdots & X_{2,p-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{n,p-1} \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

We call $\mathbf{y}$ response vector, $\mathbf{X}$ design matrix, and $\boldsymbol{\varepsilon}$ error vector. If we assume

$$E[\boldsymbol{\varepsilon}] = \mathbf{0}, \quad \text{Cov}[\boldsymbol{\varepsilon}] = \mathbf{I}_n \sigma^2,$$

then

$$E[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta}, \text{ Cov}[\mathbf{y}] = \mathbf{I}_n\sigma^2.$$

3.2 Estimation of Regression Coefficients

3.2.1 Least Squares Method

Note that the LSE of $\boldsymbol{\beta}$ is given by minimizing

$$S = \sum_{i=1}^n \varepsilon_i^2 = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}),$$

and, therefore,

$$\frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathbf{X}'\mathbf{y} + 2(\mathbf{X}'\mathbf{X})\boldsymbol{\beta}$$

which gives the following normal equation

$$(\mathbf{X}'\mathbf{X})\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}.$$

If $\mathbf{X}'\mathbf{X}$ is invertible, then

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

The *residual vector* is defined as

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}},$$

where $\hat{\mathbf{y}} = (\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_n)'$ is *fitted vector* and $\hat{Y}_i = x_i\hat{\boldsymbol{\beta}}$

As a special case, consider a simple linear regression model, then we have

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} n & \sum_{i=1}^n X_i \\ \sum_{i=1}^n X_i & \sum_{i=1}^n X_i^2 \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n X_i Y_i \end{bmatrix}$$

and

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{n \sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2} \begin{bmatrix} \sum_{i=1}^n X_i^2 & -\sum_{i=1}^n X_i \\ -\sum_{i=1}^n X_i & n \end{bmatrix}$$

3.2.2 Properties of LSE

(1) Mean and Covariance of $\hat{\boldsymbol{\beta}}$

$$E[\hat{\boldsymbol{\beta}}] = E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{y}] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$$

$$Cov[\hat{\boldsymbol{\beta}}] = Cov[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'Cov[\mathbf{y}]\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1}\sigma^2$$

(2) Gauss-Markov Theorem

Theorem : In the multiple linear regression model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $E[\boldsymbol{\varepsilon}] = \mathbf{0}$, $Cov(\boldsymbol{\varepsilon}) = \mathbf{I}_n\sigma^2$, the LSE $\hat{\boldsymbol{\beta}}$ has the minimum variance among all the unbiased and linear (in $\mathbf{y}$), i.e., $\hat{\boldsymbol{\beta}}$ is BLUE (Best Linear Unbiased Estimator).

(Proof) Let $\hat{\boldsymbol{\beta}}^*$ be any other unbiased linear estimator of $\boldsymbol{\beta}$, then for some $p \times n$ matrix $\mathbf{A}$, we must have

$$\hat{\boldsymbol{\beta}}^* = \mathbf{A}\mathbf{y}$$

and, further, for some $p \times n$ matrix D , we must have

$$\hat{\beta}^* = ((X'X)^{-1}X' + D)y$$

Now,

$$E[\hat{\beta}^*] = ((X'X)^{-1}X' + D)E[y] = ((X'X)^{-1}X' + D)X\beta = \beta + DX\beta.$$

By the unbiasedness, we have $DX = \mathbf{0}$. Now,

$$Cov(\hat{\beta}^*) = ACov(y)A' = AA'\sigma^2 = ((X'X)^{-1} + DD')\sigma^2 = Cov(\hat{\beta}) + DD'\sigma^2$$

By considering the j th diagonal element of the above equation, we have

$$Var(\hat{\beta}_j^*) = Var(\hat{\beta}_j) + \sigma^2 \sum_{i=1}^n d_{ji}^2,$$

where d_{ji}^2 is the (j, i) th component of D , and the proof is completed.

(3) Maximum Likelihood Estimator

Since the likelihood function is given by

$$\begin{aligned} L &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}\varepsilon_i^2\right) \\ &= \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} \exp\left[-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\right]. \end{aligned}$$

Therefore, maximizing L w.r.t β is equivalent to minimizing $(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$, i.e., the MLE of β is equal to the LSE of β .

(4) Useful Results

1. $\sum_{i=1}^n e_i = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{i1} - \cdots - \hat{\beta}_{p-1} X_{i,p-1}) = 0.$
2. $\sum_{i=1}^n X_{ij} e_i = 0, \quad j = 1, 2, \cdots, p-1$
3. $\sum_{i=1}^n \hat{Y}_i e_i = 0$
4. The fitted regression model always go through $(\bar{X}_1, \bar{X}_2, \cdots, \bar{X}_{p-1}, \bar{Y})$

3.3 Goodness-of-Fit and ANOVA

(1) Decomposition of Sum of Squares

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

Source	SS	d.f.	MS	F-ratio
Regression	SSR	$p-1$	$\text{MSR} = \text{SSR} / (p-1)$	$F_0 = \text{MSR} / \text{MSE}$
Error	SSE	$n-p$	$\text{MSE} = \text{SSE} / (n-p)$	
Total	SST	$n-1$		

(2) Standard Error

$$s = \sqrt{\frac{1}{n-p} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2} = \sqrt{\text{MSE}}$$

(3) Coefficient of Determination

$$R^2 = \frac{\text{SSR}}{\text{SST}}$$

(4) Test for the Significance of Regression Model

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0$$

$$H_1 : \text{At least one of } p - 1 \text{ coefficients are not } 0$$

Then, under $H_0 : \beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0$, the test statistic

$$F_0 = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} = \frac{\text{MSR}}{\text{MSE}}$$

follows an $F(p-1, n-p)$ distribution.

Ex. 3.4 (p.113)

(5) Multiple Linear Regression with Zero Intercept

(i) model : $Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i$

(ii) $\text{SST} = \sum_{i=1}^n Y_i^2$, $\text{SSR} = \sum_{i=1}^n \hat{Y}_i^2$, $\text{SSE} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$

(iii) ANOVA Table

Source	SS	d.f.	MS	F-ratio
Regression	SSR	$p-1$	$\text{MSR} = \text{SSR}/(p-1)$	$F_0 = \text{MSR}/\text{MSE}$
Error	SSE	$n-p+1$	$\text{MSE} = \text{SSE}/(n-p+1)$	
Total	SST	n		

3.4 Quadratic Forms and Distribution of Sum of Squares

3.4.1 Quadratic Forms of Sum of Squares

(1) Useful Notations

(i) summing vector : $\mathbf{1} = (1, 1, \dots, 1)^T$

(ii) J_n : $n \times n$ matrix with all components 1

(iii) properties : $\mathbf{1}\mathbf{1}' = J$, $\mathbf{1}'\mathbf{1} = n$, $J\mathbf{1} = n\mathbf{1}$, $\mathbf{1}'J = n\mathbf{1}'$, $J^2 = nJ$

(2) SST

$$\text{SST} = \sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n Y_i^2 - n(\bar{Y})^2 = \sum_{i=1}^n Y_i^2 - \frac{1}{n} \left(\sum_{i=1}^n Y_i \right)^2$$

and $\sum_{i=1}^n Y_i^2 = \mathbf{y}'\mathbf{y}$, $\sum_{i=1}^n Y_i = \mathbf{1}'\mathbf{y}$ give

$$\text{SST} = \mathbf{y}'\mathbf{y} - \frac{1}{n} \mathbf{y}'\mathbf{1}\mathbf{1}'\mathbf{y} = \mathbf{y}' \left(I - \frac{1}{n} J \right) \mathbf{y} = \mathbf{y}' T \mathbf{y},$$

$$T = I - \frac{1}{n} J.$$

(3) SSE

$$\text{SSE} = (\mathbf{y} - \hat{\mathbf{y}})'(\mathbf{y} - \hat{\mathbf{y}}) = (\mathbf{y} - X\hat{\boldsymbol{\beta}})'(\mathbf{y} - X\hat{\boldsymbol{\beta}}) = \mathbf{y}'\mathbf{y} - 2\mathbf{y}'X\hat{\boldsymbol{\beta}} + \hat{\boldsymbol{\beta}}'X'X\hat{\boldsymbol{\beta}}$$

and $\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y}$ gives

$$\text{SSE} = \mathbf{y}'(I - X(X'X)^{-1}X')\mathbf{y} = \mathbf{y}'E\mathbf{y},$$

$$E = I - X(X'X)^{-1}X'.$$

(4) SSR

Since $SSR = SST - SSE$,

$$SSR = y'Ty - y'Ey = y'(X(X'X)^{-1}X' - \frac{1}{n}J)y = y'Ry,$$

$$R = X(X'X)^{-1}X' - \frac{1}{n}J.$$

(5) Lemma : T, R , and E are idempotent.

(Proof) It is easy to prove that T and E are idempotent. Here, we prove R is idempotent. Note that

$$\begin{aligned} R \cdot R &= (X(X'X)^{-1}X' - \frac{1}{n}J)(X(X'X)^{-1}X' - \frac{1}{n}J) \\ &= X(X'X)^{-1}X' - \frac{1}{n}X(X'X)^{-1}X'J - \frac{1}{n}JX(X'X)^{-1}X' + \frac{1}{n^2}J^2. \end{aligned}$$

Now, note that

$$X'X(X'X)^{-1}X' = X',$$

and because the 1st column of X is $\mathbf{1}$, we have

$$\mathbf{1}'X(X'X)^{-1}X' = \mathbf{1}'.$$

Multiply $\mathbf{1}$ to the left side of the above equation, then

$$\mathbf{J}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' = \mathbf{J}.$$

Take the transpose of the above equation, we have

$$\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{J} = \mathbf{J}.$$

Further, $\mathbf{J}^2 = n\mathbf{J}$ implies

$$\mathbf{R} \cdot \mathbf{R} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \frac{1}{n}\mathbf{J} - \frac{1}{n}\mathbf{J} + \frac{1}{n}\mathbf{J} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \frac{1}{n}\mathbf{J} = \mathbf{R}$$

(6) Rank of $\mathbf{T}$, $\mathbf{R}$, and $\mathbf{E}$

If a matrix is idempotent, then the rank is equal to the trace of a matrix.

Also, if we use $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$, then

$$\text{tr}(\mathbf{R}) = \text{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \frac{1}{n}\mathbf{J}) = \text{tr}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) - \text{tr}(\frac{1}{n}\mathbf{J}) = \text{tr}(\mathbf{I}_p) - 1 = p - 1.$$

Also, it is easy to show that

$$\text{tr}(\mathbf{T}) = n - 1, \quad \text{tr}(\mathbf{E}) = n - p$$

3.4.2 Distribution of Sum of Squares

(1) Distribution of Sum of Squares

Theorem 3.2 Assume that $\mathbf{y} \sim N(\boldsymbol{\mu}, \mathbf{V})$. Then, the necessary and sufficient condition for $\mathbf{y}'\mathbf{A}\mathbf{y} \sim \chi^2(r, \lambda)$, where $r = r(\mathbf{A}\mathbf{V})$ and $\lambda = \frac{1}{2}\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$, is that $\mathbf{A}\mathbf{V}$ is idempotent.

As an example, apply the theorem to SSE. Since $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \mathbf{I}_n\sigma^2)$ and

$$\frac{\text{SSE}}{\sigma^2} = \mathbf{y}'\left(\frac{1}{\sigma^2}\mathbf{E}\right)\mathbf{y}$$

we need to show $\frac{1}{\sigma^2}\mathbf{E} \cdot \mathbf{I}\sigma^2$ is idempotent. It is trivial because $\mathbf{E}$ is idempotent. Now, the d.f. is the rank of $\mathbf{E}$, so that it is $(n - p)$, and the non-centrality parameter is $\lambda = \frac{1}{2}\boldsymbol{\beta}'\mathbf{X}'\left(\frac{1}{\sigma^2}\mathbf{E}\right)\mathbf{X}\boldsymbol{\beta}$. Since

$$\mathbf{X}'\mathbf{E} = \mathbf{X}'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = \mathbf{X}' - \mathbf{X}' = \mathbf{0}$$

we have $\lambda = 0$. Finally, $\text{SSE}/\sigma^2 \sim \chi^2(n - p)$. Here, we summarize the distributions of sum of squares.

1. $\text{SST}/\sigma^2 \sim \chi^2((n - 1), \frac{1}{2\sigma^2}\boldsymbol{\beta}'\mathbf{X}'(\mathbf{I} - \frac{1}{n}\mathbf{J})\mathbf{X}\boldsymbol{\beta})$
2. $\text{SSE}/\sigma^2 \sim \chi^2(n - p)$
3. $\text{SSR}/\sigma^2 \sim \chi^2((p - 1), \frac{1}{2\sigma^2}\boldsymbol{\beta}'\mathbf{X}'(\mathbf{I} - \frac{1}{n}\mathbf{J})\mathbf{X}\boldsymbol{\beta})$

(2) Expected Values of Sum of Squares

Useful fact : If $W \sim \chi^2(r, \lambda)$, then $E(W) = r + 2\lambda$

Application to sum of squares

$$(i) E[SSE] = (n - p)\sigma^2$$

$$(ii) E[SST] = (n - 1)\sigma^2 + \boldsymbol{\beta}'\mathbf{X}'(\mathbf{I} - \frac{1}{n}\mathbf{J})\mathbf{X}\boldsymbol{\beta}$$

$$(iii) E[SSR] = (p - 1)\sigma^2 + \boldsymbol{\beta}'\mathbf{X}'(\mathbf{I} - \frac{1}{n}\mathbf{J})\mathbf{X}\boldsymbol{\beta}$$

Theorem 3.3 Assume that $\mathbf{y} \sim N(\boldsymbol{\mu}, \mathbf{V})$. Then,

1. Two quadratic forms $\mathbf{y}'\mathbf{A}\mathbf{y}$ and $\mathbf{y}'\mathbf{B}\mathbf{y}$ are independent if either $\mathbf{A}\mathbf{V}\mathbf{B} = \mathbf{0}$ or $\mathbf{B}\mathbf{V}\mathbf{A} = \mathbf{0}$.
2. A quadratic form $\mathbf{y}'\mathbf{A}\mathbf{y}$ and a linear transformation $\mathbf{B}\mathbf{y}$ are independent if $\mathbf{B}\mathbf{V}\mathbf{A} = \mathbf{0}$

As an example, SSR and SSE are independent because

$$\begin{aligned} \mathbf{R} \cdot (\mathbf{I}\sigma^2) \cdot \mathbf{E} &= \sigma^2[(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \frac{1}{n}\mathbf{J})(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')] \\ &= \sigma^2[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \frac{1}{n}\mathbf{J} + \frac{1}{n}\mathbf{J}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] \\ &= \mathbf{0} \end{aligned}$$

Therefore,

$$F_0 = \frac{\frac{SSR}{\sigma^2} / (p - 1)}{\frac{SSE}{\sigma^2} / (n - p)} = \frac{MSR}{MSE}$$

follows a noncentral $F((p - 1), (n - p))$. Under $H_0 : \beta_1 = \beta_2 = \dots = \beta_{p-1} = 0$, F_0 follows a central $F((p - 1), (n - p))$.

Theorem 3.4 Let $\mathbf{y}$ be random vector with $E(\mathbf{y}) = \boldsymbol{\mu}$ and $Cov(\mathbf{y}) = \mathbf{V}$.

Then,

$$E[\mathbf{y}'\mathbf{A}\mathbf{y}] = \text{tr}(\mathbf{A}\mathbf{V}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$$

(Proof) First, note that

$$E[\mathbf{y}'\mathbf{A}\mathbf{y}] = E[\text{tr}(\mathbf{y}'\mathbf{A}\mathbf{y})] = E[\text{tr}(\mathbf{A}\mathbf{y}\mathbf{y}')] = \text{tr}(\mathbf{A}E[\mathbf{y}\mathbf{y}'])$$

and if we use $E[\mathbf{y}\mathbf{y}'] = \mathbf{V} + \boldsymbol{\mu}\boldsymbol{\mu}'$, then

$$E[\mathbf{y}'\mathbf{A}\mathbf{y}] = \text{tr}(\mathbf{A}\mathbf{V} + \mathbf{A}\boldsymbol{\mu}\boldsymbol{\mu}') = \text{tr}(\mathbf{A}\mathbf{V}) + \text{tr}(\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}) = \text{tr}(\mathbf{A}\mathbf{V}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$$

which completes the proof.

Using the Theorem 3.4, we compute $E(SSR)$. Since $\boldsymbol{\mu} = E[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta}$ and

$\mathbf{V} = I\sigma^2$, we have

$$\begin{aligned} E[\mathbf{y}'\mathbf{R}\mathbf{y}] &= \text{tr}(\sigma^2\mathbf{R}) + \boldsymbol{\beta}'\mathbf{X}'(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \frac{1}{n}\mathbf{J})\mathbf{X}\boldsymbol{\beta} \\ &= (p-1)\sigma^2 + \boldsymbol{\beta}'\mathbf{X}'(\mathbf{I} - \frac{1}{n}\mathbf{J})\mathbf{X}\boldsymbol{\beta}. \end{aligned}$$

3.5 Statistical Inference (I)

We assume that

$$\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, I_n\sigma^2).$$

3.5.1 Inference on β

(1) Distribution of $\hat{\beta}$

Since $\hat{\beta} = (X'X)^{-1}X'y$ is linear in y , it is normally distributed, i.e.,

$$\hat{\beta} \sim N(\beta, (X'X)^{-1}\sigma^2)$$

Further, the marginal distribution of the multivariate normal distribution is also normal, we have

$$\hat{\beta}_j \sim N(\beta_j, c_{j+1,j+1}\sigma^2) \quad j = 0, 1, 2, \dots, p-1,$$

where $c_{j+1,j+1}$ is the $(j+1)$ st diagonal components of $(X'X)^{-1}$.

(2) C.I. of β_j

Since the standard error of $\hat{\beta}_j$ is

$$SE(\hat{\beta}_j) = \sqrt{c_{j+1,j+1}} \cdot s$$

we have

$$t_j = \frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \quad j = 0, 1, 2, \dots, p-1$$

which follows $t(n-p)$. By using this result, the $100(1-\alpha)\%$ C.I. for β_j is

$$\hat{\beta}_j - t_{\alpha/2}(n-p)SE(\hat{\beta}_j) < \beta_j < \hat{\beta}_j + t_{\alpha/2}(n-p)SE(\hat{\beta}_j)$$

(3) Test of β_j

(i) null hypothesis : $H_0 : \beta_j = \beta_{j,0}$

(ii) test statistic : $t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{SE}(\hat{\beta}_j)} \sim t(n - p)$

null hypo.	test stat.	alternative hypo.	rejection reg.
$H_0 : \beta_j = \beta_{j,0}$	$t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{SE}(\hat{\beta}_j)}$	$H_1 : \beta_j < \beta_{j,0}$	$t_j < -t_{\alpha}(n - p)$
		$H_1 : \beta_j > \beta_{j,0}$	$t_j > t_{\alpha}(n - p)$
		$H_1 : \beta_j \neq \beta_{j,0}$	$ t_j > t_{\alpha/2}(n - p)$

Ex. 3.5 (p.122)

3.5.2 Joint Confidence Region and the Hypotheses Testing

(1) Joint Confidence Region for β

Consider a quadratic form

$$Q = (\hat{\beta} - \beta)' X' X (\hat{\beta} - \beta)$$

and note that $(\hat{\beta} - \beta) \sim N(\mathbf{0}, (X'X)^{-1}\sigma^2)$. Therefore, $Q/\sigma^2 \sim \chi^2(p)$ by Theorem 3.2. Also, if we use Theorem 3.3, then can show Q and SSE are independent. Hence,

$$\frac{\frac{Q}{\sigma^2}/p}{\frac{\text{SSE}}{\sigma^2}/(n - p)} = \frac{1}{ps^2} (\hat{\beta} - \beta)' X' X (\hat{\beta} - \beta) \sim F(p, n - p)$$

Finally,

$$\Pr \left[\frac{(\hat{\beta} - \beta)' X' X (\hat{\beta} - \beta)}{p s^2} \leq F_{\alpha}(p, n - p) \right] = 1 - \alpha$$

Also, we call

$$\{\beta : (\beta - \hat{\beta})' X' X (\beta - \hat{\beta}) \leq p s^2 F_{\alpha}(p, n - p)\}$$

as $100(1 - \alpha)\%$ *joint confidence region*.

As a special case, consider $p = 2$ (simple linear regression).

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad \hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix}, \quad X' X = \begin{bmatrix} n & \sum_{i=1}^n X_i \\ \sum_{i=1}^n X_i & \sum_{i=1}^n X_i^2 \end{bmatrix}$$

Hence,

$$\begin{aligned} & n(\beta_0 - \hat{\beta}_0)^2 + 2n(\beta_0 - \hat{\beta}_0)(\beta_1 - \hat{\beta}_1) \sum_{i=1}^n X_i + (\beta_1 - \hat{\beta}_1)^2 \sum_{i=1}^n X_i^2 \\ & \leq 2s^2 F_{\alpha}(2, n - 2) \end{aligned}$$

Ex. 3.6 (p.124)

(3) Test of β

(i) hypotheses : $H_0 : \beta = \beta_0$ vs $H_1 : \beta \neq \beta_0$

(ii) rejection region : $(\beta_0 - \hat{\beta})' X' X (\beta_0 - \hat{\beta}) > p s^2 F_{\alpha}(p, n - p)$

Fig. 3.2 (p. 125)

Ex. 3.7 (p.125)

3.5.3 Simultaneous C.I.

(1) Bonferroni Method

Consider computing simultaneous confidence intervals for g parameters

$\theta_1, \theta_2, \dots, \theta_g$. Let E_i be the event that a C.I. contains θ_i , and assume that

$1 - \alpha^*$ confidence was used in each C.I., i.e.,

$$\Pr[E_i] \geq 1 - \alpha^* \text{ or } \Pr[E_i^c] < \alpha^*$$

Now, if we let Q be the event that all the C.I.'s are correct, then we may write

$$\Pr[Q] = \Pr[\cap_{i=1}^g E_i]$$

and if we apply the Bonferroni's inequality (i.e., $\Pr[\cup_{i=1}^g E_i^c] \leq \sum_{i=1}^g \Pr[E_i^c]$),

then

$$\Pr[Q] = 1 - \Pr[\cup_{i=1}^g E_i^c] \geq 1 - \sum_{i=1}^g \Pr[E_i^c] \geq 1 - g \cdot \alpha^*$$

Therefore, the confidence of each interval should be $1 - \alpha/g$ if we want at least $1 - \alpha$ confidence for all intervals. Consequently, the $100(1 - \alpha)\%$ Bonferroni confidence interval for any β_j is

$$\hat{\beta}_j \pm t_{\alpha/2g}(n - p)SE(\hat{\beta}_j)$$

Note that $t_{\alpha/2g}$ is very large if g is large, so that the width of Bonferroni confidence interval becomes too large, and this phenomenon is a serious disadvantage of Bonferroni confidence interval.

(2) Scheffe Method

$$\hat{\beta}_j \pm \sqrt{gF_{\alpha}(g, n-p)}SE(\hat{\beta}_j)$$

(2) Maximum Modulus Method

$$\hat{\beta}_j \pm u_{\alpha}(g, n-p)SE(\hat{\beta}_j),$$

where u is a r.v. representing the maximum of the absolute of g independent t r.v.'s. with $n-p$ degrees of freedom and $u_{\alpha}(g, n-p)$ denotes the $100\alpha\%$ of the maximum modulus distribution.

Ex. 3.8 (p.128)

3.5.4 Inference on the Mean of the Response

(1) Inference on $E[Y(x_0)] = x_0'\beta$

Let $x_0 = (1, x_{01}, x_{02}, \dots, x_{0,p-1})'$ be given, and $\mu_{Y.x_0} = x_0'\beta$ be the corresponding mean of response at x_0 . We consider the statistical inference on $E[Y(x_0)] = x_0'\beta$. As a point estimator, we have

$$\hat{Y}(x_0) = x_0'\hat{\beta}$$

and its expectation is

$$E[\hat{Y}(x_0)] = E[x_0' \hat{\beta}] = x_0' \beta,$$

and its variance is

$$Var(x_0' \hat{\beta}) = x_0' Cov(\hat{\beta}) x_0 = x_0' (X' X)^{-1} x_0 \sigma^2$$

Hence, $100(1 - \alpha)\%$ C.I. for $\mu_{Y.x_0} = x_0' \beta$ becomes

$$x_0' \hat{\beta} \pm t_{\alpha/2}(n - p) \cdot s \sqrt{x_0' (X' X)^{-1} x_0}$$

We can apply this result to the C.I. for any type of linear combination of regression coefficients

$$q' \beta = q_0 \beta_0 + q_1 \beta_1 + \cdots + q_{p-1} \beta_{p-1}$$

i.e., $100(1 - \alpha)\%$ C.I. for q is

$$q' \hat{\beta} \pm t_{\alpha/2}(n - p) \cdot s \sqrt{q' (X' X)^{-1} q}$$

For example, if we want C.I. for $\beta_2 - \beta_3$ for the regression model with $p = 5$, then let $q = (0, 0, 1, -1, 0)'$. Similarly, for C.I. of $\beta_1 + \beta_3 - 2\beta_4$, we just let $q = (0, 1, 0, 1, -2)'$

(2) Inference on the New Response at x_0

100(1 - α)% C.I. for a new response $Y(x_0)$ at $x = x_0$ is

$$x_0' \hat{\beta} \pm t_{\alpha/2}(n-p) \cdot s \sqrt{1 + x_0'(X'X)^{-1}x_0}$$

Ex. 3.9 (p.130)

3.5.5 Residual Analysis

$H = X(X'X)^{-1}X'$: projection matrix, hat matrix

$\hat{y} = Hy$: fitted vector

$e = (I - H)y$: residual vector

Ex. 3.10 (p.132)

3.6 Statistical Inference (II)

3.6.1 Partial F -test

(1) Hypothesis

$$H_0 : \beta_r = \beta_{r+1} = \cdots = \beta_{p-1} = 0$$

(2) Full and Reduced Model

(i) full model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

$$SST = SSE(F) + SSR(F)$$

(ii) reduced model

Under $H_0 : \beta_r = \beta_{r+1} = \dots = \beta_{p-1} = 0$,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_{r-1} X_{i,r-1} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

$$\text{SST} = \text{SSE(R)} + \text{SSR(R)}$$

(3) Test Statistic

(i) motivation

Reject H_0 if $\text{SSR(F)} - \text{SSR(R)}$ is large

(ii) derivation

In $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, decompose

$$\mathbf{X} = [\mathbf{X}_1 \quad \mathbf{X}_2], \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$$

such that $\mathbf{X}_1$ is $n \times r$. Then,

$$\mathbf{y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$$

and the null hypothesis can be written as $H_0 : \boldsymbol{\beta}_2 = \mathbf{0}$. Note that

$$\begin{aligned} \text{SSR(F)} &= \mathbf{y}'(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \frac{1}{n}\mathbf{J})\mathbf{y} \\ \text{SSR(R)} &= \mathbf{y}'(\mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1' - \frac{1}{n}\mathbf{J})\mathbf{y} \end{aligned}$$

and therefore,

$$\text{SSR}(\text{F}) - \text{SSR}(\text{R}) = \mathbf{y}'[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' - \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1']\mathbf{y}$$

which is often called *extra sum of squares due to β_2 adjusted for β_1* , and

denoted by

$$\text{R}(\beta_2|\beta_1) = \text{SSR}(\text{F}) - \text{SSR}(\text{R}) = \text{SSE}(\text{R}) - \text{SSE}(\text{F})$$

Under $H_0 : \beta_2 = \mathbf{0}$, $\text{R}(\beta_2|\beta_1)/\sigma^2 \sim \chi^2(p-r)$, so that

$$F_0 = \frac{\text{R}(\beta_2|\beta_1)/(p-r)}{\text{SSE}(\text{F})/(n-p)} = \frac{(\text{SSE}(\text{R}) - \text{SSE}(\text{F}))/(df_R - df_F)}{\text{SSE}(\text{F})/df_F} \sim F((p-r), (n-p))$$

under $H_0 : \beta_2 = \mathbf{0}$. Hence, we reject $H_0 : \beta_2 = \mathbf{0}$ if $F > F_\alpha(p-r, n-p)$,

and we call it *partial F-test*.

Ex. 3.11 (p.136)

3.6.2 The General Linear Test

(1) Hypothesis

The general linear test can be written as

$$H_0 : \mathbf{C}\beta = \mathbf{m}, \quad H_1 : \mathbf{C}\beta \neq \mathbf{m},$$

where $\mathbf{C}$ is $q \times p$ matrix with $r(\mathbf{C}) = q$, and $\mathbf{m}$ is $q \times 1$ vector. For example,

if we want to test $H_0 : \beta_1 - \beta_2 = 0, \beta_3 + 2\beta_4 = 3$, then

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}, \quad \mathbf{m} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

Also, when we test $\beta_1 = \beta_2 = \dots = \beta_{p-1}$ which can be written as $\beta_1 - \beta_2 = 0, \beta_2 - \beta_3 = 0, \dots, \beta_{p-2} - \beta_{p-1} = 0$, and therefore, C is $(p-2) \times p$ matrix, i.e.,

$$C = \begin{bmatrix} 0 & 1 & -1 & 0 & \dots & 0 \\ 0 & 0 & 1 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & -1 \end{bmatrix}$$

and m is $(p-2)$ -dimensional vector with zeros.

(2) Test Statistic

$$F_0 = \frac{\text{SSE(R)} - \text{SSE(F)}}{q \cdot \text{MSE(F)}} \sim F(q, n - p)$$

Now, we can show that

$$\text{SSE(R)} - \text{SSE(F)} = (C\hat{\beta} - m)'[C(X'X)^{-1}C']^{-1}(C\hat{\beta} - m)$$

so that

$$F_0 = \frac{(C\hat{\beta} - m)'[C(X'X)^{-1}C']^{-1}(C\hat{\beta} - m)}{q \cdot \text{MSE(F)}}$$

Ex. 3.13 (p.140)

3.6.3 Inclusion of More Covariates

(1) Augmented Model

So far we considered $E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$. Now we consider an augmented model

$$G : E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} = (\mathbf{X} \ \mathbf{Z}) \begin{pmatrix} \boldsymbol{\beta} \\ \boldsymbol{\gamma} \end{pmatrix} \equiv \mathbf{W}\boldsymbol{\delta},$$

where $\mathbf{X}$ is $n \times p$ matrix with $r(\mathbf{X}) = p$, $\mathbf{Z}$ is $n \times k$ matrix with $r(\mathbf{Z}) = k$,

and $\mathbf{W}$ is $n \times (p + k)$ matrix with $r(\mathbf{W}) = p + k$.

(2) LSE of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$

To compute the LSEs of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$, we minimize

$$\begin{aligned} \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} &= (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\boldsymbol{\gamma})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\boldsymbol{\gamma}) \\ &= \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} - 2\boldsymbol{\gamma}'\mathbf{Z}'\mathbf{y} + 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\gamma}'\mathbf{Z}'\mathbf{Z}\boldsymbol{\gamma} \end{aligned}$$

and the derivative w.r.t. $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ gives

$$-2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{Z}\hat{\boldsymbol{\gamma}}_G + 2\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}_G = \mathbf{0}$$

$$-2\mathbf{Z}'\mathbf{y} + 2\mathbf{Z}'\mathbf{X}\hat{\boldsymbol{\beta}}_G + 2\mathbf{Z}'\mathbf{Z}\hat{\boldsymbol{\gamma}}_G = \mathbf{0}$$

From the 1st eq., we have

$$\hat{\boldsymbol{\beta}}_G = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}_G)$$

and plug this into the 2nd eq., we have

$$\mathbf{Z}'\mathbf{Z}\hat{\boldsymbol{\gamma}}_G = \mathbf{Z}'\mathbf{y} - \mathbf{Z}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}_G)$$

and so that

$$\mathbf{Z}'[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{Z}\hat{\gamma}_G = \mathbf{Z}'[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{y}$$

Now, let $\mathbf{R} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, then

$$\hat{\gamma}_G = (\mathbf{Z}'\mathbf{R}\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{R}\mathbf{y}$$

(2) Test on $H_0 : \gamma = \mathbf{0}$

First, we note that $SSE(R) = \mathbf{y}'\mathbf{R}\mathbf{y}$ and let $\mathbf{H}_G = \mathbf{W}(\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'$ be the hat matrix under the full model, and $\mathbf{R}_G = \mathbf{I} - \mathbf{W}(\mathbf{W}'\mathbf{W})^{-1}\mathbf{W}'$. Then,

$$\begin{aligned} SSE(F) &= \mathbf{y}'\mathbf{R}_G\mathbf{y} \\ &= (\mathbf{y} - \mathbf{W}\hat{\delta}_G)'(\mathbf{y} - \mathbf{W}\hat{\delta}_G) \\ &= (\mathbf{y} - \mathbf{X}\hat{\beta}_G - \mathbf{Z}\hat{\gamma}_G)'(\mathbf{y} - \mathbf{X}\hat{\beta}_G - \mathbf{Z}\hat{\gamma}_G) \end{aligned}$$

Since

$$\begin{aligned} \mathbf{y} - \mathbf{X}\hat{\beta}_G - \mathbf{Z}\hat{\gamma}_G &= \mathbf{y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{y} - \mathbf{Z}\hat{\gamma}_G) - \mathbf{Z}\hat{\gamma}_G \\ &= [\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'](\mathbf{y} - \mathbf{Z}\hat{\gamma}_G) \\ &= \mathbf{R}(\mathbf{y} - \mathbf{Z}\hat{\gamma}_G) \end{aligned}$$

we have

$$\begin{aligned}
SSE(F) &= \mathbf{y}'\mathbf{R}_G\mathbf{y} \\
&= (\mathbf{y} - \mathbf{Z}\hat{\gamma}_G)'\mathbf{R}(\mathbf{y} - \mathbf{Z}\hat{\gamma}_G) \\
&= \mathbf{y}'\mathbf{R}\mathbf{y} - 2\hat{\gamma}_G'\mathbf{Z}'\mathbf{R}\mathbf{y} + \hat{\gamma}_G'\mathbf{Z}'\mathbf{R}\mathbf{Z}\hat{\gamma}_G \\
&= \mathbf{y}'\mathbf{R}\mathbf{y} - \hat{\gamma}_G'\mathbf{Z}'\mathbf{R}\mathbf{y} - \hat{\gamma}_G'(\mathbf{Z}'\mathbf{R}\mathbf{y} - \mathbf{Z}'\mathbf{R}\mathbf{Z}\hat{\gamma}_G) \\
&= \mathbf{y}'\mathbf{R}\mathbf{y} - \hat{\gamma}_G'\mathbf{Z}'\mathbf{R}\mathbf{y} \\
&= SSE(R) - \hat{\gamma}_G'\mathbf{Z}'\mathbf{R}\mathbf{y}
\end{aligned}$$

Therefore, the F -test statistic for testing $H_0 : \gamma = \mathbf{0}$ is

$$\begin{aligned}
F &= \frac{(SSE(R) - SSE(F))/(df_R - df_F)}{SSE(F)/df_F} \\
&= \frac{\hat{\gamma}_G'\mathbf{Z}'\mathbf{R}\mathbf{y}/((n - p - k) - (n - p))}{(\mathbf{y}'\mathbf{R}\mathbf{y} - \hat{\gamma}_G'\mathbf{Z}'\mathbf{R}\mathbf{y})/(n - p - k)} \\
&= \frac{\hat{\gamma}_G'\mathbf{Z}'\mathbf{R}\mathbf{y}/k}{(\mathbf{y}'\mathbf{R}\mathbf{y} - \hat{\gamma}_G'\mathbf{Z}'\mathbf{R}\mathbf{y})/(n - p - k)}
\end{aligned}$$

which follows $F(k, n - p - k)$ distribution under $H_0 : \gamma = \mathbf{0}$.

3.7 Lack of Fit Test

(1) Hypothesis

$$H_0 : E[Y_i] = \mathbf{X}_i'\boldsymbol{\beta}, \quad i = 1, 2, \dots, n$$

(2) Test Statistic

Assume that $Y_{il}(i = 1, 2, \dots, m; l = 1, 2, \dots, n_i)$ is the l th replication at x_i . Then,

$$(Y_{il} - \hat{Y}_i) = (Y_{il} - \bar{Y}_i) + (\bar{Y}_i - \hat{Y}_i) \quad i = 1, 2, \dots, m; l = 1, 2, \dots, n_i$$

from which we have

$$\sum_{i=1}^m \sum_{l=1}^{n_i} (Y_{il} - \hat{Y}_i)^2 = \sum_{i=1}^m \sum_{l=1}^{n_i} (Y_{il} - \bar{Y}_i)^2 + \sum_{i=1}^m n_i (\bar{Y}_i - \hat{Y}_i)^2$$

because $\sum_{l=1}^{n_i} (Y_{il} - \bar{Y}_i) = 0$, i.e., we decompose SSE as follows;

$$SSE = SSPE + SSLF,$$

where SSPE and SSLF denote *pure error sum of squares* and *lack – of – fit sum of squares*, respectively. Also, the corresponding degrees of freedoms are

$$(n - p) = (n - m) + (m - p).$$

Further, mean squares are obtained if we divide sum of squares by its d.f.,

i.e.,

$$MS_{PE} = \frac{\sum_{i=1}^m \sum_{l=1}^{n_i} (Y_{il} - \bar{Y}_i)^2}{\sum_{i=1}^m (n_i - 1)}$$

is called *pure error mean squares*, and

$$MS_{LF} = \frac{\sum_{i=1}^m n_i (\bar{Y}_i - \hat{Y}_i)^2}{m - p}$$

is called *lack of fit mean squares*. Finally, the test statistic for testing H_0 :

$$E[Y_i] = \mathbf{X}_i' \boldsymbol{\beta} \text{ is}$$

$$F_{LF} = \frac{MS_{LF}}{MS_{PE}} \sim F_{\alpha}(m - p, n - m),$$

and the ANOVA table is given by the followings;

source	SS	d.f.	MS	F-ratio
Regression	SSR	$p - 1$	MSR	$F_{LF} = MS_{LF}/MS_{PE}$
Error	SSE	$n - p$	MSE	
Lack-of-Fit	SSLF	$m - p$	MS_{LF}	
Pure Error	SSPE	$n - m$	MS_{PE}	
Total	SST	$n - 1$		

Ex. 3.14 (p.146)

3.8 Miscellanea

3.8.1 Standardization of Regression Coefficients

Let

$$S_{jj} = \sum_{i=1}^n (X_{ij} - \bar{X}_j)^2, \quad S_{YY} = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

and let

$$\hat{\beta}_j^* = \sqrt{\frac{S_{ij}}{S_{YY}}} \hat{\beta}_j, \quad j = 1, 2, \dots, p - 1$$

which is called *standardized regression coefficient*. Further, let

$$Y_i^* = \frac{Y_i - \bar{Y}}{\sqrt{S_{YY}}}, \quad W_{ij} = \frac{X_{ij} - \bar{X}_j}{\sqrt{S_{jj}}}, \quad i = 1, 2, \dots, n; \quad j = 1, 2, \dots, p - 1$$

Then,

$$(\mathbf{W}'\mathbf{W})\hat{\beta}^* = \mathbf{W}'\mathbf{y}^*$$

Note that

$$\mathbf{W}'\mathbf{W} = \begin{bmatrix} 1 & r_{12} & r_{13} & \cdots & r_{1,p-1} \\ r_{12} & 1 & r_{23} & \cdots & r_{2,p-1} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ r_{1,p-1} & r_{2,p-1} & r_{3,p-1} & \cdots & 1 \end{bmatrix}$$

and

$$\{\mathbf{W}'\mathbf{W}\}_{jl} = \sum_{i=1}^n W_{ij}W_{il} = \frac{\sum_{i=1}^n (X_{ij} - \bar{X}_j)(X_{il} - \bar{X}_l)}{\sqrt{S_{jj}S_{ll}}} = r_{jl}$$

Also,

$$\mathbf{W}'\mathbf{y}^* = \begin{bmatrix} r_{Y1} \\ r_{Y2} \\ \cdots \\ r_{Y,p-1} \end{bmatrix}$$

Ex. 3.15 (p.150)

3.8.2 Multicollinearity

Assume that we have two covariates and if we use standardization of covariates, then

$$\mathbf{W}'\mathbf{W} = \begin{bmatrix} 1 & r_{12} \\ r_{12} & 1 \end{bmatrix}$$

and its determinant is $|\mathbf{W}'\mathbf{W}| = 1 - r_{12}^2$ and the inverse matrix is

$$(\mathbf{W}'\mathbf{W})^{-1} = \frac{1}{1 - r_{12}^2} \begin{bmatrix} 1 & -r_{12} \\ -r_{12} & 1 \end{bmatrix}$$

Hence, if two covariates are correlated, i.e., r_{12} is close to ± 1 , then $\mathbf{W}'\mathbf{W}$ is close to singular matrix. Therefore, the corresponding estimates of regression coefficients become very unstable. In general, if some covariates are highly correlated, then the resulting estimates of regression coefficients are not reliable. In this case, we say there exists *multicollinearity* between covariates.

3.8.3 Geometric Interpretation of LSE

$$\| \mathbf{y} \|^2 = \| \hat{\mathbf{y}} \|^2 + \| \mathbf{e} \|^2$$

3.8.4 Relationship with the Multivariate Normal Distribution

Let $\mathbf{X} = (X_1, \dots, X_{p-1})$ be $p - 1$ dimensional covariates and Y be a response. We assume that $(\mathbf{X}, Y) \sim N_p$. Let $f(\mathbf{X}, \boldsymbol{\beta}) = \beta_0 + \beta_1 X_1 + \dots + \beta_{p-1} X_{p-1}$. Then,

$$\begin{aligned} f(\mathbf{X}, \boldsymbol{\beta}) &= E(Y|\mathbf{X}) \\ &= E(Y) + \text{Cov}(Y, \mathbf{X})\text{Cov}(\mathbf{X})^{-1}(\mathbf{X} - E(\mathbf{X})) \\ &= E(Y) - \text{Cov}(Y, \mathbf{X})\text{Cov}(\mathbf{X})^{-1}E(\mathbf{X}) + \text{Cov}(Y, \mathbf{X})\text{Cov}(\mathbf{X})^{-1}\mathbf{X} \\ &\equiv \beta_0 + (\beta_1, \dots, \beta_{p-1})'\mathbf{X} \end{aligned}$$

i.e.,

$$\begin{aligned}\beta_0 &= E(Y) - Cov(Y, \mathbf{X})Cov(\mathbf{X})^{-1}E(\mathbf{X}) \\ (\beta_1, \dots, \beta_{p-1})' &= Cov(\mathbf{X})^{-1}Cov(\mathbf{X}, Y)\end{aligned}$$

As a special case, in a simple linear regression model,

$$\begin{aligned}\hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1\bar{X} \\ \hat{\beta}_1 &= \frac{1}{n}\sum(X_i - \bar{X})(Y_i - \bar{Y}) / \frac{1}{n}\sum(X_i - \bar{X})^2.\end{aligned}$$

Chapter 4. Regression Diagnostics

4.1 Introduction

regression diagnostics (model checking)

(i) linearity : $E(Y) = \sum_{j=0}^{p-1} \beta_j X_j$

cf. nonlinear regression, generalized linear model, nonparametric regression

(ii) distribution of error terms : $\varepsilon_i \sim N(0, \sigma^2)$

cf. Durbin-Watson test, robust regression

(iii) multicollinearity

$X'X$ is close to singular (ill-conditioned matrix)

cf. ridge regression

(iv) influential observations

4.2 Residual

(i) Def.

$$e_i = Y_i - \hat{Y}_i, \quad i = 1, 2, \dots, n, \text{ i.e., } \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}.$$

(ii) Properties

$$E(\mathbf{e}) = \mathbf{0}, \quad \text{Cov}(\mathbf{e}) = (\mathbf{I} - \mathbf{H})\sigma^2,$$

where $H = X(X'X)^{-1}X'$ is *hat matrix*.

$$\text{Var}(e_i) = (1 - h_{ii})\sigma^2, \quad \text{Cov}(e_i, e_j) = -h_{ij}\sigma^2 \quad (i \neq j).$$

Note that

$$h_{ij} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_j,$$

where $\mathbf{x}_i = (1 \ x_{i1} \ x_{i2} \ \cdots \ x_{i,p-1})'$ is the i th row vector of $\mathbf{X}$. Therefore, the correlation coefficient of e_i and e_j becomes

$$\text{Corr}(e_i, e_j) = -\frac{h_{ij}}{\sqrt{1 - h_{ii}}\sqrt{1 - h_{jj}}} \quad (i \neq j)$$

4.2.1 Internally Studentized Residual

The standardized form of the residual e_i is

$$\frac{e_i - 0}{\sigma\sqrt{1 - h_{ii}}},$$

however, σ is unknown, and we replace it by its estimator s , then

$$r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}$$

which is called *internally Studentized residual*. It is known that

$$r_i^2/(n - p) \sim \text{Beta}\left(\frac{1}{2}, (n - p - 1)/2\right).$$

4.2.2 Externally Studentized Residual

externally Studentized (cross validatory, jackknife) residual :

$$r_i^* = \frac{e_i}{s_{(i)}\sqrt{1-h_{ii}}} \sim t(n-p-1),$$

where $s_{(i)}^2$ is the unbiased estimator of σ^2 based on $(n-1)$ observations after deleting the i th observation, and it can be shown that

$$s_{(i)}^2 = s^2 \cdot \frac{n-p-r_i^2}{n-p-1}$$

and

$$r_i^* = r_i \sqrt{\frac{n-p-1}{n-p-r_i^2}}$$

4.3 Leverage

(1) Def. of Leverage

$h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$: i th leverage, the distance between $\mathbf{x}_i$ and $\bar{\mathbf{x}}$.

In simple linear regression case,

$$h_{ij} = \begin{pmatrix} 1 & x_i \end{pmatrix} \frac{1}{S_{XX}} \begin{pmatrix} \sum x_i^2/n & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix} \begin{pmatrix} 1 \\ x_j \end{pmatrix} = \frac{1}{n} + \frac{(x_i - \bar{x})(x_j - \bar{x})}{S_{XX}}$$

and

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{XX}}$$

(2) Properties of Leverage

(i) $\sum_{i=1}^n h_{ii} = p$

(Proof)

$$\sum_{i=1}^n h_{ii} = \text{tr}(\mathbf{H}) = \text{tr}[\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'] = \text{tr}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}] = \text{tr}(\mathbf{I}_p) = p$$

(ii) $\frac{1}{n} \leq h_{ii} \leq 1$

(Proof) Since $\mathbf{H}$ is idempotent, we have

$$h_{ii} = \sum_{j=1}^n h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$$

which means that $h_{ii} \geq h_{ii}^2$, so that $h_{ii} \leq 1$. Now, to show $h_{ii} \geq \frac{1}{n}$, let $\tilde{\mathbf{X}}$ be the centered matrix of $\mathbf{X}$, then can show that

$$h_{ii} = \frac{1}{n} + (\mathbf{x}_i - \bar{\mathbf{x}})'(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}(\mathbf{x}_i - \bar{\mathbf{x}})$$

and therefore, $h_{ii} \geq \frac{1}{n}$.

(iii) $h_{ij}^2 \leq h_{ii}h_{jj}$

(Proof) Since $\mathbf{X}'\mathbf{X}$ is p.d., there exists an upper triangular matrix $\mathbf{R}$ s.t.

$\mathbf{X}'\mathbf{X} = \mathbf{R}'\mathbf{R}$. Let $\mathbf{a} = (\mathbf{R}^{-1})'\mathbf{x}_i$, $\mathbf{b} = (\mathbf{R}^{-1})'\mathbf{x}_j$, then

$$h_{ij} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_j = \mathbf{x}_i'(\mathbf{R}'\mathbf{R})^{-1}\mathbf{x}_j = \mathbf{a}'\mathbf{b}.$$

By the Cauchy-Schwarz inequality, i.e., $(\mathbf{a}'\mathbf{b})^2 \leq (\mathbf{a}'\mathbf{a})(\mathbf{b}'\mathbf{b})$, we have

$$\begin{aligned} h_{ij}^2 &\leq [((\mathbf{R}^{-1})'\mathbf{x}_i)'(\mathbf{R}^{-1})'\mathbf{x}_i] [((\mathbf{R}^{-1})'\mathbf{x}_j)'(\mathbf{R}^{-1})'\mathbf{x}_j] \\ &= \mathbf{x}_i'(\mathbf{R}'\mathbf{R})^{-1}\mathbf{x}_i\mathbf{x}_j'(\mathbf{R}'\mathbf{R})^{-1}\mathbf{x}_j = h_{ii}h_{jj} \end{aligned}$$

(3) High-Leverage Point

An observation $\mathbf{x}_i$ is called a *high – leverage point* if the corresponding h_{ii} is large. Also, note that

$$\hat{y}_i = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \sum_{j=1}^n \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_j y_j = h_{ii}y_i + \sum_{j \neq i} h_{ij}y_j$$

i.e., $\hat{y}_i$ is largely affected by y_i and if h_{ii} is also large, then $\hat{y}_i$ is affected a lot by x_i . To determine whether $\mathbf{x}_i$ is a high-leverage point or not, the rule of thumb cutoff value is $2p/n$ because the average value of h_{ii} is p/n .

4.4 Influential Observations

(1) Def. : An observation is called *influential observation* if it has a large effect on estimators $\hat{\beta}$, s^2 , etc.

(2) Remarks

(i) Influential on what?

(ii) "singly influential observation" vs "set of influential observations"

(iii) "masking effect" vs "swamping phenomenon"

Fig. 4.2 (p. 172)

4.5 Influence Measures

(1) Four Approaches of Influence Measure

(i) Deletion Method

To measure the influence of the i th observation on $\hat{\beta}$, the *deletion method* computes $\hat{\beta} - \hat{\beta}_{(i)}$, where $\hat{\beta}_{(i)}$ is the LSE of β based on $n - 1$ observations after deleting the i th observation. In fact, $\hat{\beta}_{(i)} = (\mathbf{X}'_{(i)} \mathbf{X}_{(i)})^{-1} \mathbf{X}'_{(i)} \mathbf{y}_{(i)}$, where $\mathbf{X}_{(i)}$ is $(n - 1) \times p$ matrix deleting the i th row vector of $\mathbf{X}$ and $\mathbf{y}_{(i)}$ is $(n - 1)$ -vector deleting y_i from $\mathbf{y}$.

(ii) Infinitesimal Perturbation Method

Assume that $y_j \sim N(x'_j \beta, \sigma^2/w_j)$, $j = 1, \dots, n$, and let

$$w_j = \begin{cases} 1, & j = 1, \dots, i-1, i+1, \dots, n \\ w, & j = i \end{cases}$$

where $0 \leq w \leq 1$. Under this assumption, let the LSE of β be $\hat{\beta}_w$. As a special case, $\hat{\beta}_1 = \hat{\beta}$ and $\hat{\beta}_0 = \hat{\beta}_{(i)}$. The *infinitesimal perturbation method* is seeing the performance of $\hat{\beta}_w$ when w goes from 1 to 0.

(iii) Local Influence

Let $\mathbf{D} = \text{diag}(e_1, \dots, e_n)$ be $n \times n$ diagonal matrix, and $\mathbf{l}$ be n -vector

with $\|l\| = 1$. Also, let

$$C_l = l' K l,$$

where $K = DHD$. Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ be eigenvalues of K and $l_1, l_2, \dots, l_n$ be corresponding eigenvectors. Then, λ_1 contains the information on the influence of observations on $\hat{\beta}$, and this method is called the *local influence*.

(iv) Replacement Measure

To evaluate the influence of the i th observation, replace y_i by $\hat{y}_i$ instead of deleting the i th observation, and this approach is called the *replacement method*.

4.5.1 Influence Measures by the Deletion Method

(1) Cook's Distance

The *Cook's distance* (Cook, 1977) for the i th observation is

$$C_i = \frac{1}{p} (\hat{\beta} - \hat{\beta}_{(i)})' \text{Cov}(\hat{\beta})^{-1} (\hat{\beta} - \hat{\beta}_{(i)})$$

and can show that

$$\hat{\beta} - \hat{\beta}_{(i)} = (X'X)^{-1} x_i e_i / (1 - h_{ii}).$$

Therefore,

$$C_i = \frac{1}{p\sigma^2} \cdot \frac{e_i^2 h_{ii}}{(1 - h_{ii})^2}$$

If σ^2 is unknown, then replace it by its estimator s^2 . Also,

$$C_i = \frac{1}{p} \cdot \frac{r_i^2 h_{ii}}{(1 - h_{ii})},$$

where r_i is the internally Studentized residual. Note that it is an increasing function of residual and leverage. As a reference value for C_i , Kim and Storer (1996) suggested $C_i \geq 3.67/(n - p)$.

Now, for the influence of set of observations, let $K = \{i_1, \dots, i_k\}$ be an index set of size k . Then, we consider $\hat{\beta} - \hat{\beta}_{(K)}$, where $\hat{\beta}_{(K)} = (\mathbf{X}'_{(K)} \mathbf{X}_{(K)})^{-1} \mathbf{X}'_{(K)} \mathbf{y}_{(K)}$. The Cook's distance for k observations in K is

$$C_K = \frac{1}{p} (\hat{\beta} - \hat{\beta}_{(K)})' \text{Cov}(\hat{\beta})^{-1} (\hat{\beta} - \hat{\beta}_{(K)})$$

can show that

$$\hat{\beta} - \hat{\beta}_{(K)} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'_K (\mathbf{I} - \mathbf{H}_K)^{-1} \mathbf{e}_K,$$

where $\mathbf{H}_K = \mathbf{X}_K (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'_K$, and therefore,

$$C_K = \frac{1}{p\sigma^2} \mathbf{e}'_K (\mathbf{I} - \mathbf{H}_K)^{-1} \mathbf{H}_K (\mathbf{I} - \mathbf{H}_K)^{-1} \mathbf{e}_K.$$

(2) Andrews-Pregibon Statistic

Let $\mathbf{X}^* = (\mathbf{X}, \mathbf{y})$ be $n \times (p + 1)$ augmented matrix. Then, the *Andrews – Pregibon statistic* (Andrews and Pregibon, 1978) for the i th observation is

$$AP_i = \det[\mathbf{X}_{(i)}^{*'} \mathbf{X}_{(i)}^*] / \det[\mathbf{X}^{*'} \mathbf{X}^*]$$

and if we use

$$\det[\mathbf{X}^{*'}\mathbf{X}^*] = \mathbf{e}'\mathbf{e}\det[\mathbf{X}'\mathbf{X}]$$

then

$$AP_i = (1 - h_{ii})\{1 - e_i^2 / (1 - h_{ii})\mathbf{e}'\mathbf{e}\} = 1 - h_{ii}^*,$$

where $h_{ii}^* = \mathbf{x}_i^{*'}(\mathbf{X}^{*'}\mathbf{X}^*)^{-1}\mathbf{x}_i^*$. For the influence of k observations in K ,

$$\begin{aligned} AP_K &= \det[\mathbf{X}_K^{*'}\mathbf{X}_K^*] \det[\mathbf{X}^{*'}\mathbf{X}^*] \\ &= \det[\mathbf{I} - \mathbf{H}_K] \{1 - \mathbf{e}_K'(\mathbf{I} - \mathbf{H}_K)^{-1}\mathbf{e}_K / \mathbf{e}'\mathbf{e}\} \\ &= \det[\mathbf{I} - \mathbf{H}_K^*], \end{aligned}$$

where $\mathbf{H}_K^* = \mathbf{X}_K^*(\mathbf{X}^{*'}\mathbf{X}^*)^{-1}\mathbf{X}_K^{*}$.

(3) DFBETAS

Belsley, Kuh and Welsch (1980) suggested the influence of the i th observation on $\hat{\beta}_j$, $j = 0, 1, \dots, p - 1$ instead of $\hat{\beta}$. First, consider $p \times n$ matrix $\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \equiv (c_{ij})$. Then,

$$\hat{\beta}_j - \hat{\beta}_{j(i)} = \frac{c_{ji}e_i}{1 - h_{ii}}$$

and

$$\text{Cov}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\sigma^2, \quad \mathbf{C}\mathbf{C}' = (\mathbf{X}'\mathbf{X})^{-1}.$$

Therefore,

$$\text{Var}(\hat{\beta}_j) = \sigma^2 \sum_{l=1}^n C_{jl}^2$$

and the influence of the i th observation on $\hat{\beta}_j$, $j = 0, 1, \dots, p-1$ is

$$\frac{\hat{\beta}_j - \hat{\beta}_{j(i)}}{\sqrt{\text{Var}(\hat{\beta}_j)}} = \frac{\hat{\beta}_j - \hat{\beta}_{j(i)}}{\sigma \sqrt{(\mathbf{X}'\mathbf{X})_{jj}^{-1}}}$$

Since σ is unknown, Belsley, Kuh and Welsch (1980) suggested using $s_{(i)}$,

and call it *DFBETAS* (Difference in betas), i.e.,

$$DFBETAS_{j,i} = \frac{\hat{\beta}_j - \hat{\beta}_{j(i)}}{s_{(i)} \sqrt{(\mathbf{X}'\mathbf{X})_{jj}^{-1}}} = \frac{c_{ji}}{\sqrt{\sum_{l=1}^n c_{jl}^2}} \cdot \frac{e_i}{s_{(i)}(1 - h_{ii})}$$

Now, for the influence of k observations in $K = \{i_1, \dots, i_k\}$, let

$\mathbf{c}_{jk} = (c_{ji_1}, \dots, c_{ji_k})'$, then

$$\hat{\beta}_j - \hat{\beta}_{j(K)} = \mathbf{c}_{jk}'(\mathbf{I} - \mathbf{H}_K)^{-1} \mathbf{e}_K$$

which gives

$$DFBETAS_{j,K} = \frac{\hat{\beta}_j - \hat{\beta}_{j(K)}}{s_{(K)} \sqrt{(\mathbf{X}'\mathbf{X})_{jj}^{-1}}} = \mathbf{c}_{jk}'(\mathbf{I} - \mathbf{H}_K)^{-1} \mathbf{e}_K / s_{(K)} \sqrt{\sum_{l=1}^n c_{jl}^2}.$$

(4) DFFITS

The influence of the i th observation on the fitted value $\hat{y}_i$ can be measured

by $\hat{y}_i - \hat{y}_{i(i)}$, where $\hat{y}_{i(i)} = \mathbf{x}_i' \hat{\boldsymbol{\beta}}_{(i)}$. Now, the standardized form is

$$|\hat{y}_i - \hat{y}_{i(i)}| / \text{SE}(\hat{y}_i) = \frac{|\mathbf{x}_i'(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})|}{\sigma \sqrt{h_{ii}}}.$$

Belsley, Kuh and Welsch (1980) suggested DFFITS (Difference in Standardized Fits) as

$$DFFITS_i = \frac{|\mathbf{x}'_i(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})|}{s_{(i)}\sqrt{h_{ii}}} = \frac{1}{s_{(i)}} \cdot \frac{|e_i|\sqrt{h_{ii}}}{1 - h_{ii}} = \frac{|r_i^*|\sqrt{h_{ii}}}{\sqrt{1 - h_{ii}}}$$

For the influence of k observations in K , we have

$$DFFITS_K = \frac{1}{s_{(K)}} \cdot \frac{\mathbf{x}'_i(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'_K(\mathbf{I} - \mathbf{H}_K)^{-1}\mathbf{e}_K}{\sqrt{h_{ii}}}.$$

(5) COVRATIO

The influence of the i th observation on $Cov(\hat{\boldsymbol{\beta}})$, we compare $Cov(\hat{\boldsymbol{\beta}}_{(i)}) = \sigma^2(\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1}$ and $Cov(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. To do this, Belsley, Kuh and Welsch (1980) suggested

$$COVRATIO_i = \det[s_{(i)}^2(\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1}] / \det[s^2(\mathbf{X}'\mathbf{X})^{-1}]$$

If we use

$$\det[\mathbf{X}'_{(i)}\mathbf{X}_{(i)}] = (1 - h_{ii})\det[\mathbf{X}'\mathbf{X}]$$

then

$$COVRATIO_i = 1 / \left\{ \frac{n - p - 1}{n - p} + \frac{r_i^{*2}}{n - p} \right\}^p (1 - h_{ii})$$

For the reference value for $COVRATIO_i$, they suggested $|COVRATIO_i - 1| \geq 3p/n$. For the influence of k observations in K , we have

$$COVRATIO_K = \det[s_{(K)}^2(\mathbf{X}'_{(K)}\mathbf{X}_{(K)})^{-1}] / \det[s^2(\mathbf{X}'\mathbf{X})^{-1}]$$

and we may use

$$\det[\mathbf{X}'_{(K)}\mathbf{X}_{(K)}] = \det[\mathbf{I} - \mathbf{H}_K]\det[\mathbf{X}'\mathbf{X}].$$

4.5.2 Other Approaches

(1) Influence Curve

$$\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_w = \frac{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i(1-w)e_i}{\{1 - (1-w)h_{ii}\}}$$

$$\Delta\hat{\boldsymbol{\beta}}_w = \frac{\partial\hat{\boldsymbol{\beta}}_w}{\partial w} = \frac{-(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_ie_i}{\{1 - (1-w)h_{ii}\}^2}$$

$$\Delta\hat{\boldsymbol{\beta}}_1 = -(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_ie_i : \textit{influence curve}$$

(2) replacement measure

(i) Idea : replace y_i by $\hat{y}_i$

Let $\mathbf{y}_{r(i)} = (y_1, \dots, y_{i-1}, \hat{y}_i, y_{i+1}, \dots, y_n)'$, then the corresponding LSE of $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}}_{r(i)} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}_{r(i)}$. Kim(1996) proposed an influence measure for the i th observation as

$$R_i = \frac{1}{p}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{r(i)})'Cov(\hat{\boldsymbol{\beta}})^{-1}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{r(i)})$$

which is called *replacement measure*. If we let $\mathbf{E}_i = \text{diag}(0, \dots, 0, 1, 0, \dots, 0)$, then

$$\mathbf{y}_{r(i)} = (\mathbf{I} - \mathbf{E}_i)\mathbf{y} + \mathbf{E}_i\hat{\mathbf{y}}$$

$$\begin{aligned}
\hat{\beta}_{r(i)} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'[(\mathbf{I} - \mathbf{E}_i)\mathbf{y} + \mathbf{E}_i\hat{\mathbf{y}}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'[\mathbf{y} - \mathbf{E}_i(\mathbf{y} - \hat{\mathbf{y}})] \\
&= \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{E}_i\mathbf{e}
\end{aligned}$$

Hence,

$$\begin{aligned}
R_i &= \frac{1}{p\sigma^2}\mathbf{e}'\mathbf{E}_i\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{E}_i\mathbf{e} \\
&= \frac{1}{p\sigma^2}\mathbf{e}'\mathbf{E}_i\mathbf{H}\mathbf{E}_i\mathbf{e} \\
&= \frac{1}{p\sigma^2}e_i^2h_{ii}
\end{aligned}$$

If σ^2 is unknown, use its estimator s^2 .

For the influence of k observations in K , we have

$$\mathbf{y}_{r(K)} = (\mathbf{I} - \mathbf{E}_K)\mathbf{y} + \mathbf{E}_K\hat{\mathbf{y}}$$

then,

$$\begin{aligned}
\hat{\beta}_{r(K)} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'[(\mathbf{I} - \mathbf{E}_K)\mathbf{y} + \mathbf{E}_K\hat{\mathbf{y}}] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'[\mathbf{y} - \mathbf{E}_K(\mathbf{y} - \hat{\mathbf{y}})] \\
&= \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{E}_K\mathbf{e}
\end{aligned}$$

Therefore,

$$\begin{aligned}
R_K &= \frac{1}{p} (\hat{\beta} - \hat{\beta}_{r(K)})' \text{Cov}(\hat{\beta})^{-1} (\hat{\beta} - \hat{\beta}_{r(K)}) \\
&= \frac{1}{p\sigma^2} e' E_K X (X'X)^{-1} (X'X) (X'X)^{-1} X' E_K e \\
&= \frac{1}{p\sigma^2} e' E_K H E_K e \\
&= \frac{1}{p\sigma^2} \left\{ \sum_{i \in K} e_i^2 h_{ii} + 2 \sum_{i \neq j, i, j \in K} e_i e_j h_{ij} \right\}
\end{aligned}$$

Ex. 4.1 (p.182)

4.6 Test for Outliers

To test whether k observations out of n observations are outliers, consider a *outlier shift model*

$$y = X\beta + Z\gamma + \varepsilon,$$

where

$$Z = \begin{pmatrix} \mathbf{0} \\ I_k \end{pmatrix}$$

is $n \times k$ matrix. Now, the null hypothesis for the existence of outliers can be written as $H_0 : \gamma = \mathbf{0}$. Hence, under H_0 , the outlier shift model reduces to

$$y = X\beta + \varepsilon$$

Now, we decompose the hat matrix as

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \mathbf{H}_{22} \end{pmatrix},$$

where $\mathbf{H}_{11}$ is $(n - k) \times (n - k)$, $\mathbf{H}_{12}$ is $(n - k) \times k$, $\mathbf{H}_{21}$ is $k \times (n - k)$, $\mathbf{H}_{22}$ is $k \times k$ matrices. Also, we decompose $\mathbf{e} = (\mathbf{e}'_1 \ \mathbf{e}'_2)'$.

To derive the test statistic for testing $H_0 : \gamma = \mathbf{0}$, note that

$$\begin{aligned} \hat{\gamma} &= [\mathbf{Z}'(\mathbf{I}_n - \mathbf{H})\mathbf{Z}]^{-1}\mathbf{Z}'(\mathbf{I}_n - \mathbf{H})\mathbf{y} \\ &= \left[(\mathbf{0}, \mathbf{I}_k)(\mathbf{I}_n - \mathbf{H}) \begin{pmatrix} \mathbf{0} \\ \mathbf{I}_k \end{pmatrix} \right]^{-1} (\mathbf{0}, \mathbf{I}_k) \begin{pmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{pmatrix} \\ &= (\mathbf{I}_k - \mathbf{H}_{22})^{-1}\mathbf{e}_2 \end{aligned}$$

and

$$\hat{\gamma}'\mathbf{Z}'(\mathbf{I}_n - \mathbf{H})\mathbf{y} = \hat{\gamma}'(\mathbf{0}, \mathbf{I}_k)\mathbf{e} = \hat{\gamma}'\mathbf{e}_2 = \mathbf{e}'_2(\mathbf{I}_k - \mathbf{H}_{22})^{-1}\mathbf{e}_2.$$

Hence, the test statistic becomes

$$F = \frac{\mathbf{e}'_2(\mathbf{I}_k - \mathbf{H}_{22})^{-1}\mathbf{e}_2/k}{[RSS - \mathbf{e}'_2(\mathbf{I}_k - \mathbf{H}_{22})^{-1}\mathbf{e}_2]/(n - p - k)}.$$

As a special case, consider $k = 1$ case. Since

$$\mathbf{e}'_2(\mathbf{I}_k - \mathbf{H}_{22})^{-1}\mathbf{e}_2 = \frac{e_i^2}{1 - h_{ii}}$$

the F -statistic reduces to

$$\frac{(n - p - 1)e_i^2}{(1 - h_{ii})SSE - e_i^2} \sim F(1, n - p - 1)$$

Note that the internally Studentized residual is

$$r_i^2 = \frac{e_i^2}{s^2(1 - h_{ii})}$$

and the F –statistic is

$$t_i^2 = \frac{(n - p - 1)r_i^2}{n - p - r_i^2}$$

which is the square of the externally Studentized residual.

4.7 Multicollinearity

(1) Definition

If covariates are highly correlated, then we say there exists *multicollinearity*.

If multicollinearity exists, then there are serious problems in the statistical inference on regression coefficients.

$$\mathbf{W}'\mathbf{W} = (\mathbf{W}'\mathbf{W})^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

i.e, estimates of variance of regression coefficients are 1. But, if

$$\mathbf{W}'\mathbf{W} = \begin{bmatrix} 1 & 0.975 \\ 0.975 & 1 \end{bmatrix}, \quad (\mathbf{W}'\mathbf{W})^{-1} = \begin{bmatrix} 20.253 & -19.747 \\ -19.747 & 20.253 \end{bmatrix}$$

then estimates of variance of regression coefficients are 20.253.

(2) Methods of Checking the Existence of Multicollinearity

(i) *variance inflation factor* (VIF_k)

$$VIF_k = \frac{1}{1 - R_k^2}, \quad k = 1, 2, \dots, p-1,$$

where R_k^2 is the coefficient of determination when regressing k th covariate on other covariates. Multicollinearity might exist if $\max_k VIF_k > 10$.

(ii) eigenvalues of $\mathbf{X}'\mathbf{X}$.

Let $d_1 \geq d_2 \geq \dots \geq d_p \geq 0$ be singular values of $\mathbf{X}$. Define

$$\kappa(\mathbf{X}) = d_1/d_p$$

be *condition number*. It is clear that $\kappa(\mathbf{X}) = (\lambda_1/\lambda_p)^{1/2}$, where λ_1 and λ_p are the largest and smallest eigenvalues of $\mathbf{X}'\mathbf{X}$, respectively.

Ex. 4.2 (p.188)

4.8 Autocorrelation and Durbin-Watson Test

(1) Autocorrelation Model

Consider *first – order autoregressive model*

$$\varepsilon_t = \rho\varepsilon_{t-1} + \delta_t, \quad |\rho| < 1,$$

where δ_t is iid with $E(\delta_t) = 0$, $\text{Var}(\delta_t) = \sigma_\delta^2$, and ρ is *first – order autocorrelation coefficient*

of $\varepsilon_1, \dots, \varepsilon_n$. Now, note that

$$\begin{aligned}
 \varepsilon_t &= \rho(\rho\varepsilon_{t-2} + \delta_{t-1}) + \delta_t \\
 &= \rho^2(\rho\varepsilon_{t-3} + \delta_{t-2}) + \rho\delta_{t-1} + \delta_t \\
 &= \delta_t + \rho\delta_{t-1} + \rho^2\delta_{t-2} + \dots \\
 &= \sum_{j=0}^{\infty} \rho^j \delta_{t-j}
 \end{aligned}$$

so that

$$E(\varepsilon_t) = \sum_{j=0}^{\infty} \rho^j E(\delta_{t-j}) = 0$$

and

$$Cov(\varepsilon_t, \varepsilon_{t-l}) = \frac{\rho^l \sigma_{\delta}^2}{1 - \rho^2}$$

We can compute the variance of ε_t if we let $l = 0$ in the above equation,

$$Var(\varepsilon_t) = \frac{\sigma_{\delta}^2}{1 - \rho^2}$$

Further,

$$Corr(\varepsilon_t, \varepsilon_{t-l}) = \frac{Cov(\varepsilon_t, \varepsilon_{t-l})}{\sqrt{Var(\varepsilon_t)Var(\varepsilon_{t-l})}} = \rho^l, \quad l \neq 0$$

(2) Durbin-Watson Test

$$\begin{aligned}
E(\varepsilon_t - \varepsilon_{t-1})^2 &= E(\varepsilon_t^2 + \varepsilon_{t-1}^2 - 2\varepsilon_t\varepsilon_{t-1}) \\
&= E(\varepsilon_t^2) + E(\varepsilon_{t-1}^2) - 2E(\varepsilon_t\varepsilon_{t-1}) \\
&= 2\sigma_\varepsilon^2 - 2\frac{\rho\sigma_\varepsilon^2}{1-\rho^2}
\end{aligned}$$

Therefore, we have

$$\rho = 0, \quad E(\varepsilon_t - \varepsilon_{t-1})^2 = 2\sigma_\varepsilon^2$$

$$\rho > 0, \quad E(\varepsilon_t - \varepsilon_{t-1})^2 < 2\sigma_\varepsilon^2$$

$$\rho < 0, \quad E(\varepsilon_t - \varepsilon_{t-1})^2 > 2\sigma_\varepsilon^2$$

If we replace ε_t by e_t , we obtain the Durbin-Watson test statistic :

$$D = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2}$$

We see that if n is large, then

$$D \approx 2(1 - \hat{\rho}),$$

where

$$\hat{\rho} = \frac{\sum_{t=2}^n e_t e_{t-1}}{\sum_{t=2}^n e_{t-1}^2}$$

Hence, if D is close to 2, then $\rho \approx 0$, and if D is close to 0, then there are positive autocorrelation. On the other hand, if D is close to 4, then there are negative autocorrelation.

The Durbin-Watson test has the following set-up ;

hypotheses : $H_0 : \rho = 0$ vs $H_1 : \rho > 0$

$D < d_L$: Reject H_0 .

$D > d_U$: Do not reject H_0 .

$d_L \leq D \leq d_U$: Cannot decide

Ex. 4.3 (p.192)

(3) Transformation Methods

Recall that

$$Y_t = \beta_0 + \beta_1 X_t + \varepsilon_t, \quad \varepsilon_t = \rho \varepsilon_{t-1} + \delta_t,$$

where δ_t 's are independent and $N(0, \sigma_\delta^2)$. Let $Y'_t = Y_t - \rho Y_{t-1}$, then

$$\begin{aligned} Y'_t &= Y_t - \rho Y_{t-1} \\ &= (\beta_0 + \beta_1 X_t + \varepsilon_t) - \rho (\beta_0 + \beta_1 X_{t-1} + \varepsilon_{t-1}) \\ &= \beta_0(1 - \rho) + \beta_1(X_t - \rho X_{t-1}) + (\varepsilon_t - \rho \varepsilon_{t-1}) \\ &= \beta'_0 + \beta_1 X'_t + \delta_t \end{aligned}$$

In the above equation, we need to estimate ρ , and there are 3 methods to estimate ρ : (1) Cochrane-Orcutt method, (2) Hildreth-Lu method, and (3) difference method.